An NLI Faithfulness Reward Added Nothing to LoRA Fine-Tuning of Qwen3 and T5Gemma 2 in Arabic and Malay Question Answering: A Controlled Null Result with a Length-Matched Analysis

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Abstract

Natural language inference (NLI) entailment scores are proposed as training rewards for faithfulness, but the reward's contribution is rarely separated from the machinery around it. We test a leave-one-out REINFORCE objective with an NLI-entailment reward for low-rank adaptation (LoRA) of Qwen3-0.6B-Base and T5Gemma 2 270M-270M on Arabic (XQuAD) and Malay (Belebele) closed-passage question answering, with five seeds, a frozen reference, cross-entropy training and two controls that remove the reward's information (zero weight, shuffled reward). Fine-tuning raises F1 and exact match sharply in every cell. The reward adds nothing reliable: its effect on the NLI score is at most 0.012; nine of twelve contrasts with cross-entropy and the controls include zero, and the three that do not, all in Qwen3 on Arabic, are not reproduced by an independent NLI scorer (**+0.002**) and disappear at matched answer length (**+0.003**, 95% CI **−0.004** to **+0.009**). The same analysis shows that most of the NLI drop after fine-tuning is a change in answer length. Training logs suggest why: cross-entropy memorizes the 470 training examples, and the policy-gradient term is inactive in 80–100% of logged steps in the second half of training. A legacy mT5-base baseline, which does not memorize, shows the opposite: the reward raises the Arabic NLI score by **+0.60** while exact match falls and answers grow six-fold, consistent with over-optimization. An NLI reward can thus do nothing or do harm, and the score alone cannot tell which; controls, length-matched analysis and reward-independent metrics are needed.

Keywords: NLI reward, faithfulness, reward over-optimization, null result, low-rank adaptation, multilingual question answering

1 Introduction

Retrieval-Augmented Generation (RAG) (Lewis et al, 2020b) couples a retrieval component with a generative language model to ground answers in externally retrieved evidence, and has become a default architecture for knowledge-intensive natural language processing. Despite this grounding, RAG systems remain susceptible to hallucination: answers that are fluent and plausible but not supported by the retrieved context (Ji et al, 2023). We study this generation-stage failure in isolation: every example supplies the gold passage directly (closed-passage QA), so we evaluate training-time interventions on the generator's faithfulness to context it has already been given, not the retrieval step of a full RAG pipeline (Section 6).

The problem is compounded outside English, consistent with documented disparities in NLP resources across languages (Magueresse et al, 2020). Arabic and Malay are useful test cases because they differ substantially from each other and from English, and both are constrained here to a small (470-example) fine-tuning set typical of what a research group might collect for a specific downstream task. We use "low-resource" to mean this fine-tuning-data-scarce regime, not a claim about either language's overall NLP resource status.

A common recipe for improving faithfulness during training is to add a reward from a natural-language-inference (NLI) model, scored on the model's own samples, and to train with a policy-gradient estimator. Combined with parameter-efficient fine-tuning, in particular Low-Rank Adaptation (LoRA) (Hu et al, 2021) and its variants (QLoRA (Dettmers et al, 2023), AdaLoRA (Zhang et al, 2023), DoRA (Liu et al, 2024), VeRA (Kopiczko et al, 2024)), it is cheap enough for groups without large-scale compute. But such experiments differ from a cross-entropy baseline in more than the reward: they add sampling, an extra scoring pass, a different memory profile and often different engineering settings (in our earlier study, activation checkpointing). Unless the configuration is controlled, an effect attributed to the reward may belong to any of these.

What this paper does. We ask whether such a reward helps when it is tested on two modern small language models under a controlled protocol. The backbones are Qwen3-0.6B-Base (Qwen Team, 2025), a decoder-only model, and T5Gemma 2 270M-270M (Zhang et al, 2025b), an encoder-decoder model. Both are adapted with LoRA on Arabic (XQuAD) and Malay (Belebele) closed-passage question answering, with five seeds per cell. Besides a frozen reference and a cross-entropy baseline, every cell includes two controls that run the full sampling, scoring and policy-gradient machinery but remove the information in the reward: one sets the reward weight to zero, the other shuffles the rewards across the batch. Held-out answers are decoded in evaluation mode with the memory-saving switch off, and every answer of every run is saved, so that the NLI score can be analysed against answer length.

What we find. Fine-tuning itself matters a great deal: it raises F1 and exact match far above the frozen models in every cell. The NLI reward does not. Its effect on the NLI faithfulness score is at most 0.012 in any cell, nine of the twelve contrasts with cross-entropy training and with the two controls include zero, and the three that do not are all in one cell (Qwen3, Arabic) and are not reproduced by an independent NLI scorer. When answer length is held fixed, that remaining difference disappears.

The training logs offer an explanation: with 470 training examples and 12 epochs, cross-entropy training memorizes the training set, the sampled answers then carry no reward variance, and the policy-gradient term is close to zero in most of the second half of training. An earlier mT5-base study of the same objective, which we keep as a legacy baseline, behaves differently. There the reward did raise the Arabic NLI score by +0.60, but by turning three-word answers into sixteen-word sentences while exact match fell, which we read as over-optimization of the score. Whether an NLI reward does nothing or does harm therefore depends on the training regime, and neither outcome is visible in the NLI score alone.

Provenance. The first version of this work used mT5-base and reported that the reward harmed LoRA and DoRA. Zero-weight and shuffled-reward controls and an audit showed that this was an evaluation artifact of gradient checkpointing; after correction, the corrected mT5 study (162 launcher jobs, 228 run records, eight seeds per cell) gave the large Arabic effect just described (Appendices A and B). The first study's 188 runs are kept in Appendix C. The controlled study reported in the main text is a new, larger-scale rerun on two different backbones, with the corrected path built in from the start.

Contributions.

1. **A controlled null result for an NLI faithfulness reward on two modern backbones.** With five seeds, a frozen reference and two controls that remove the reward's information, the reward changes the NLI faithfulness score by at most 0.012 in any cell; once answer length is held fixed, no contrast with a control has an interval excluding zero (Sections 4.3 and 4.5).
2. **A length-matched analysis of faithfulness.** Within answer-length bins, and with length-bin fixed effects, the only nominally significant reward effect (Qwen3, Arabic, +0.011) falls to +0.003 (95% CI −0.004 to +0.009). The same analysis shows that most of the drop in NLI score after fine-tuning is an answer-length effect, because fine-tuning turns 13-word generations into three- or four-word spans (Section 4.5).
3. **A candidate explanation for the null.** Cross-entropy training memorizes the small training set and the policy-gradient term becomes inactive; an mT5 baseline that does not memorize shows the opposite behaviour, a large reward effect that comes with much longer answers and lower exact match (Sections 4.6 and 4.7).
4. **A reproducible protocol and release.** A corrected runner with per-example predictions for all 84 runs, the length-matched analysis script and the analysis tables are released (Section 7). The legacy mT5 study, its audit of six implementation defects and its blinded human validation are in the appendices.

Two scope statements apply throughout. First, faithfulness is an automatic NLI entailment score that is also the reward, so we report changes in an *NLI faithfulness score*, supported by exact match, F1 and an independent second scorer, and we do not claim human-judged faithfulness gains or losses for the new backbones. Second, Arabic (extractive XQuAD) and Malay (multiple-choice-derived Belebele) differ in task format as well as language, so Arabic/Malay differences are benchmark-language differences.

The remainder of the paper is organized as follows. Section 2 reviews related work; Section 3 gives the experimental setup; Section 4 reports the results; Section 5 discusses what the evidence does and does not support; Sections 6 to 8 cover limitations, reproducibility and broader impact; and Section 9 concludes. The appendices hold the legacy mT5 study, the audit and the first study, and the human validation.

2 Related Work

2.1 Retrieval-Augmented Generation and Hallucination

RAG (Lewis et al, 2020b) combines a learned retriever with a sequence-to-sequence generator, marginalizing over retrieved documents, building on work showing retrieval can be integrated into pre-training (Guu et al, 2020) and that dense passage representations outperform sparse lexical retrieval for open-domain QA (Karpukhin et al, 2020). Fusion-in-Decoder (Izacard and Grave, 2021) showed encoding passages independently and fusing only in the decoder scales to large retrieval sets, an architectural choice later generalized across RAG systems (see Gao et al (2023b)). Retrieval improves factual grounding but does not eliminate hallucination: a generator can still ignore, misread, or override retrieved context. Ji et al (2023) distinguish *intrinsic* hallucination (contradicting the source) from *extrinsic* (unverifiable from the source), mapping directly onto RAG, where the retrieved passage is the source. This paper's faithfulness score is a proxy for support by the retrieved passage: it estimates whether the generated answer is entailed by the context, independent of world-factual correctness. Entailment is not specific to hallucination. It rewards answers that paraphrase or copy the context, penalizes extrinsic content only when the classifier recognizes it as unsupported, and says nothing about whether the answer addresses the question, which is why we report exact match and F1 beside it.

2.2 Parameter-Efficient Fine-Tuning

LoRA (Hu et al, 2021) freezes the pretrained backbone and injects trainable low-rank decomposition matrices into selected weight matrices, reducing trainable parameters by orders of magnitude while retaining competitive performance. QLoRA (Dettmers et al, 2023) combines LoRA with 4-bit (NF4) quantization of the frozen backbone, reducing memory footprint. AdaLoRA (Zhang et al, 2023) replaces LoRA's fixed rank with an adaptive budget, parameterizing the update via SVD and pruning less-important singular values during training. DoRA (Liu et al, 2024) decomposes the pretrained weight into magnitude and direction, applying LoRA only to the directional update, narrowing the learning-capacity gap to full fine-tuning. VeRA (Kopiczko et al, 2024) takes the opposite efficiency direction, sharing one pair of frozen random low-rank matrices across all layers and learning only small per-layer scaling vectors.

LoRA sits within a broader PEFT family including adapter modules (Houlsby et al, 2019), prefix-tuning (Li and Liang, 2021), prompt tuning (Lester et al, 2021), BitFit (Zaken et al, 2022), and $(IA)^3$ (Liu et al, 2022). We restrict our comparison to the LoRA family because, unlike these alternatives, LoRA-style updates merge into

the base weights after training, adding no inference-time latency, relevant where generation latency is already a concern. The primary experiments of this paper use plain LoRA, because the reward acts on the training objective and not on the adapter. The legacy mT5 study (Appendix B) compares LoRA, QLoRA and DoRA, and the first study (Appendix C) also AdaLoRA and VeRA. Initialization-based (PiSSA (Meng et al, 2024)), scaling-based (rsLoRA (Kalajdzievski, 2023)) and optimization-based (LoRA+ (Hayou et al, 2024)) variants are not studied. Prior comparisons among these variants are concentrated in English, general-purpose instruction-tuning, or commonsense-reasoning benchmarks; we are not aware of a controlled test of how a fixed faithfulness-oriented training objective behaves when added to LoRA fine-tuning in Arabic and Malay.

2.3 NLI-Based Faithfulness Evaluation

Maynez et al (2020) showed via human evaluation that abstractive summarization models hallucinate unfaithful content at substantial rates, and that textual entailment correlates with human faithfulness judgments better than overlap metrics like BLEU (Papineni et al, 2002), ROUGE (Lin, 2004), or embedding-similarity metrics like BERTScore (Zhang et al, 2020a), which reward surface similarity rather than source grounding. Entailment-based faithfulness metrics use an NLI model to test whether generated text is entailed by its source: SummaC (Laban et al, 2022) proposes sentence-aggregated scoring; QA-based alternatives (Fabbri et al, 2022) and atomic-fact decomposition (Min et al, 2023) target long, multi-sentence outputs better suited than the short QA answers evaluated here; RAGAS (Es et al, 2023) packages a similar NLI-based score into a standard RAG evaluation suite without a training-time mitigation component; and more recent metrics move beyond a single off-the-shelf classifier (TRUE (Honovich et al, 2022) finds NLI- and QA-based scoring strongest; AlignScore (Zha et al, 2023) trains one alignment function across seven pooled tasks; MiniCheck (Tang et al, 2024) targets cheap, deployable fact-checking at GPT-4-level quality, the same cost concern that motivates parameter-efficient adaptation here). We use a single fine-tuned NLI classifier rather than one of these newer trained metrics because our premise–hypothesis formulation is simpler than the multi-task alignment problem they target; adopting one as a drop-in replacement for F_{faith} is a natural extension of the independent re-scoring check that we run with a second NLI model (Section 4.4). A multilingual-specific point this study surfaces (Section 3.4) is that the NLI model itself must have multilingual coverage, since an English-only classifier applied to non-English pairs produces scores reflecting its inability to parse the input rather than genuine faithfulness; we use a mDeBERTa-v3-base multilingual NLI model (He et al, 2021) fine-tuned on XNLI (Conneau et al, 2018) (including Arabic) and English MNLI, whose pretraining corpus (CC100) covers both Arabic and Malay, enabling cross-lingual transfer to Malay despite it not being among XNLI's fine-tuning languages.

2.4 Training-Time Faithfulness Interventions for RAG

Closer to this paper's core contribution is a growing line of work that intervenes on faithfulness during training rather than only at inference time. Self-RAG (Asai et al, 2024) trains a language model to emit explicit reflection tokens that critique its own generations against retrieved passages, unifying retrieval-on-demand and self-critique as next-token prediction objectives learned jointly with generation. RAGTruth (Niu et al, 2024) takes a complementary, data-centric approach, contributing a large human-annotated corpus of word-level RAG hallucinations for fine-tuning smaller detection models. More recently, SSFO (Tang et al, 2025) constructs preference pairs by contrasting a model's own outputs generated with and without retrieved context, then applies Direct Preference Optimization (Rafailov et al, 2023), a reward-model-free alternative to RLHF, to align full-model faithfulness without human-annotated labels, closer to this paper's training-time, reward-driven framing, but a full-model objective evaluated beyond English only in zero-shot transfer (Spanish XQuAD and Chinese DuReader). F^2RL (Wang et al, 2024) likewise trains a generator with faithfulness reward models by reinforcement learning, for counterspeech generation. All of these share this paper's premise that faithfulness is best addressed as a training-time objective. They vary the objective, the data or the detector, however, not the parameter-efficient adapter used to train the model, and we do not claim that faithfulness training has never been combined with LoRA or evaluated outside English. This paper's composite loss (Equation 1) is comparatively simple by design: rather than a critique vocabulary or separate detector, it folds an NLI-entailment penalty directly into a rank-constrained LoRA adapter's generation loss, cheap enough to apply per-language where training a Self-RAG- or RAGTruth-style model from scratch would be impractical. That simplicity does not make it easy to evaluate: the effect of such a reward is entangled with the sampling and scoring path and with engineering settings around it, which is the problem this paper's controls address. Our policy-gradient estimator (Equation D2) is positioned as an efficiency-oriented, multilingual complement to these larger-scale English-centric interventions.

Positioned against this landscape: Section 2.3's inference-time metrics (SummaC, QAFactEval, TRUE, RAGAS) operate on frozen outputs, intervening on neither training nor adaptation method; the training-time interventions above (Self-RAG, RAGTruth, SSFO, F^2RL) intervene on training but do not treat the adaptation *method* as an experimental factor. The gap this paper addresses is narrower than a first application of faithfulness training to PEFT or to non-English text: it is a controlled attribution analysis of one NLI-based faithfulness reward added to LoRA fine-tuning of two small backbones in two lower-resource languages, with a frozen reference and with zero-weight and shuffled-reward controls that separate the reward's contribution from the configuration around it (Section 4).

Table 1 summarizes the comparison along the dimensions that matter for this paper's question. SSFO is the closest in spirit: it also trains the generator on a faithfulness signal derived from the model itself, but it learns from preference pairs through a reward-model-free objective, adapts the full model and tests languages beyond English only by zero-shot transfer. Our setting differs in using an external NLI scorer as an explicit reward, in testing the reward against controls that remove its information.

Table 1 Training-time faithfulness interventions, as described in the cited works and in the text above. We make no claim about which of these works report controls that remove the signal's information.

Work	Faithfulness signal	What is trained	Languages
Self-RAG	Reflection tokens predicted jointly with the answer	Language model with critique vocabulary	English
RAGTruth	Human word-level hallucination labels	Smaller detection models	English
SSFO	Preference pairs from outputs with and without context; DPO	Full model	English; zero-shot Spanish and Chinese
F^2RL	Faithfulness reward models; reinforcement learning	Counterspeech generator	English
This paper	External NLI entailment reward; leave-one-out REINFORCE	LoRA (mT5 baseline: also QLoRA, DoRA)	Arabic, Malay

We do not claim to outperform any of these systems: the studies differ in task, model and metric, and we ran none of them.

2.5 Reward Over-Optimization

Optimizing a proxy reward can raise the proxy while a gold objective stalls or falls. Gao et al. (Gao et al, 2023a) measured this with synthetic gold preferences: they optimized policies against proxy reward models by best-of-n sampling and by reinforcement learning, and found that the gold reward first rises and then falls with the distance moved from the initial policy, with functional forms that depend on the reward-model size. Skalse et al. (Skalse et al, 2022) formalized when optimizing a proxy cannot be guaranteed to help the true reward, and Amodei et al. (Amodei et al, 2016) listed gaming of a misspecified reward among the practical problems of safe learning. Our study is a small controlled instance of this phenomenon, and it differs from that literature in four respects. The proxy is an NLI entailment classifier that is also the headline evaluation metric, not a learned preference model scored against a larger one. The policies are small language models (0.27 to 0.6 billion parameters in the main study, 0.58 billion for the mT5 baseline) adapted with low-rank updates, not large instruction-tuned models. The comparison includes controls that keep the sampling and scoring machinery but remove the reward's dependence on the answer, an independent second NLI scorer and, for the legacy mT5 adapter, blinded human judgments. We use one reward weight and one training length, so we do not trace a proxy-versus-gold curve or fit a scaling law. What the study adds is a pair of contrasting outcomes under the same objective: on the two modern backbones the reward is inert once the policy has memorized its training set, and on mT5, which does not memorize, it changes the form of the output by a large amount without improving the reference-based scores.

2.6 Multilingual Question-Answering Benchmarks

XQuAD (Artetxe et al, 2020) extends SQuAD 1.1 (Rajpurkar et al, 2016) to eleven languages including Arabic via professional translation, preserving the original extractive-QA format: short, verbatim answer spans lifted from the context. Belebele (Bandarkar et al, 2024) is a parallel multiple-choice reading-comprehension benchmark spanning 122 language variants including Malay (`zsm_Latn`), built on FLORES-200 passages (NLLB Team et al, 2022); unlike XQuAD, its gold answers are full-sentence paraphrases among four presented options rather than extractive spans, which matters for how each dataset is adapted into a free-text generation task (Section 3.1). We use XQuAD rather than the similar MLQA (Lewis et al, 2020a) for its wider adoption as the LoRA/PEFT-comparison literature's default Arabic benchmark. Arabic-native alternatives to XQuAD's translated spans exist (TyDi QA (Clark et al, 2020), ARCD (Mozannar et al, 2019)) and are a natural target for replicating this study's findings on non-translated, non-Modern-Standard-Arabic text (Habash, 2010), a distributional mismatch this paper does not control for. On the Malay side, this study evaluates a single benchmark; broader Southeast Asian coverage is emerging through efforts such as SEACrowd (Lovenia et al, 2024).

The main experiments use two small, recent backbones that a single research group can fine-tune on one GPU: Qwen3-0.6B-Base (Qwen Team, 2025), a decoder-only model, and T5Gemma 2 270M-270M (Zhang et al, 2025b), a 2025 encoder-decoder adapted from Gemma 3 (Gemma Team, 2025), in the spirit of recent encoder-decoder-versus-decoder-only comparisons (Zhang et al, 2025a; Jin et al, 2025). The earlier mT5-base (Xue et al, 2021) study, a massively multilingual text-to-text Transformer (Raffel et al, 2020; Vaswani et al, 2017) pretrained on 101 languages, is kept as a legacy baseline. The LoRA-variant papers were themselves evaluated primarily on decoder-only or encoder-only backbones (GPT-2/GPT-3/LLaMA for LoRA and QLoRA; RoBERTa/DeBERTa for AdaLoRA and VeRA), with limited encoder-decoder evaluation and none on mT5 (Hu et al, 2021; Dettmers et al, 2023; Zhang et al, 2023; Liu et al, 2024; Kopiczko et al, 2024). Whether findings transfer to larger-scale or instruction-tuned multilingual LLMs such as Aya (Üstün et al, 2024) remains untested (Section 6); the LoRA family is one slice of a fast-growing PEFT literature (Han et al, 2024).

3 Experimental Setup

3.1 Datasets and Task Formulation

We evaluate on the Arabic subset of XQuAD (`xquad.ar`) and the Malay subset of Belebele (`zsm_Latn`). For each language we reserve the first 470 examples for fine-tuning and evaluate on every remaining example: 720 held-out examples for Arabic and 430 for Malay. The split is fixed across the entire study (the same partition for every backbone, condition and seed), so the training seeds vary initialization and stochastic training, not which examples are used for training versus evaluation.

The split is constructed by index position, not by grouping on source passage, so a context passage can cross the boundary. We checked this directly: of XQuAD-ar's 240

unique passages exactly 1 appears in both slices (4 of 720 held-out examples, 0.6%); of Belebele-zsm's 488 unique passages, none do. The primary study keeps all 720 Arabic items and stores a flag for the four overlapping ones; excluding them changes every condition mean of the NLI score by at most 0.001 (for example Qwen3, condition B: 0.1445 with and 0.1452 without). The legacy mT5 study excluded them (716 items).

For Arabic, each example provides a context passage, a question and a short extractive gold answer (for example a number or short noun phrase), following XQuAD/SQuAD conventions. For Malay, Belebele's native format is multiple choice: a passage, a question, four answer options and the index of the correct option. To use it as a free-text generation and faithfulness task, we treat the text of the correct option as the gold target and include all four options in the prompt, after the passage, so that input truncation removes passage text rather than the question or the options. Exposing the options matters: an earlier formulation without them asked the model to recall a sentence it had never been shown and produced a depressed, nearly flat faithfulness signal in every configuration. Any multiple-choice benchmark repurposed as free-text generation needs the same treatment.

The two benchmarks differ in task format (extractive span versus option text placed in the prompt) as well as in language, so every Arabic-versus-Malay contrast below is a contrast between benchmark-language conditions, not a clean language effect. Prompts are `Question:{q}\nContext:{passage}\n[Options: (1)...(4)\n]Answer:` for Qwen3 and `question:{q} context:{passage}[ options: (1)...(4)]` for the encoder-decoder models, with the passage truncated to fit 512 prompt tokens; 2–5% of prompts are affected.

3.2 Backbones and Conditions

The two main backbones are Qwen3-0.6B-Base (Qwen Team, 2025), a decoder-only model, and T5Gemma 2 270M-270M (`google/t5gemma-2-270m-270m`) (Zhang et al, 2025b), an encoder-decoder. Both are adapted with LoRA (Hu et al, 2021) of rank r=8, α=16 and dropout 0.05 on the query, key, value and output projections and the gated feed-forward block (`gate_proj`, `up_proj`, `down_proj`) of the text modules; T5Gemma 2's vision modules are not adapted. Training uses bfloat16. Each backbone, language and seed is run under five conditions:

- **A (frozen):** the pretrained model with no adaptation, decoded greedily. It is deterministic and run once per backbone and language.
- **B (cross-entropy):** LoRA trained with token-level cross-entropy on the gold answer.
- **C (composite):** identical to B except that the objective is

$$\mathcal{L} = \mathcal{L}_{\mathrm{CE}} + \lambda_1 \mathcal{L}_{\mathrm{PG}}, \qquad \lambda_1 = 0.3, \tag{1}$$

  where $\mathcal{L}_{\mathrm{PG}}$ is a leave-one-out REINFORCE estimator (Ahmadian et al, 2024) of the negative expected NLI faithfulness score of the model's own sampled answers (Section 3.3); it plays the role of the baseline in self-critical training (Rennie et al, 2017), with the baseline taken from the other samples of the same prompt.

- **$C_{\lambda=0}$:** the same code path as C, including sampling, scoring and re-scoring, with λ_1=0. It isolates the effect of the machinery from the reward.
- **C_{shuf}:** like C, but the rewards of the batch are rotated across prompts before the advantages are formed, which preserves their distribution and removes their dependence on the sampled answer.

An effect attributable to the reward must therefore exceed what $C_{\lambda=0}$ and C_{shuf} produce. The earlier mT5-base study of the same objective is kept as a legacy baseline and was not re-run (Appendices A and B).

3.3 Training Procedure and Estimator

All trainable conditions are fine-tuned for 12 epochs with batch size 4 (1,416 optimizer steps) using AdamW at 2×10^{-4}, 5% linear warm-up followed by linear decay, and gradient clipping at 1.0. The study uses five seeds (42, 123, 2026, 7, 99). No validation split is held out, no checkpoint is selected and no hyperparameter is tuned on the held-out data: every number is from the model state at the end of the fixed schedule. The 470-example training set and the 12-epoch schedule are inherited from the legacy protocol, so that the backbones are comparable with it.

For conditions C, $C_{\lambda=0}$ and C_{shuf} each step first computes the cross-entropy loss in training mode (dropout on), then switches the model to evaluation mode for the policy branch: it samples K=4 answers per prompt at temperature 1.0 from the full softmax (no nucleus or top-k truncation, key–value cache on), stops each answer at the end-of-sequence token or at the first token containing a newline, scores the answer text with the reward model, forms the leave-one-out advantage $A_{ik} = R_{ik} - \frac{1}{K-1}\sum_{j\neq k} R_{ij}$, and re-scores the sampled tokens in a teacher-forced pass with the same model state, so that the scored distribution equals the sampled one. The policy term is $\mathcal{L}_{\mathrm{PG}} = -\,\mathrm{mean}_{i,k}\left(A_{ik}\sum_t \log \pi(y_{ikt} \mid x_i, y_{ik<t})\right)$ and covers exactly the tokens that were sampled, including the stop token. The reward is the entailment probability of the sampled answer given the *raw passage* as premise (Section 3.4); the premise is never the decoded, truncated prompt. A shared sanitizer replaces non-finite logits identically in sampling and scoring. Held-out answers are always decoded greedily (`max_new_tokens`= 40) in evaluation mode with the cache on, and the runner asserts that the model is in evaluation mode and that gradient checkpointing is off. Gradient checkpointing was not used in any reported run. Appendix D gives the derivation of the estimator and the legacy audit (Appendix A) the defects it avoids.

Run inventory and environment. The study comprises 2 backbones × 2 languages × (4 trained conditions × 5 seeds + 1 frozen run) = 84 runs, each writing a run record, every held-out prediction with its example identifier, and a training log. Runs were distributed over four NVIDIA H200 NVL GPUs with PyTorch, Transformers 5.5.0 and PEFT, with up to five processes sharing a GPU. Twelve Qwen3 runs (conditions C, $C_{\lambda=0}$ and C_{shuf}; Arabic seeds 2026 and 99, Malay seeds 42 and 99) first failed with CUDA out-of-memory errors because too many processes shared one GPU; they were restarted from scratch with identical settings and no partial result was used. During the restart, the $C_{\lambda=0}$ run of those four seed-language cells was trained by two concurrent processes, and the output records are those of whichever finished last. The

two executions of the same seed and condition differ by 0.003 to 0.012 in faithfulness and 0.007 to 0.021 in F1 (Section 4.1), which gives a direct measure of run-to-run nondeterminism.

3.4 Faithfulness Metric

F_{faith} is computed by feeding the retrieved context (the passage) as the premise and the generated answer as the hypothesis to a natural language inference classifier, and taking the softmax probability assigned to the *entailment* class, the same zero-shot NLI-as-classifier recipe validated for efficient, scalable text classification by Laurer et al (2024), whose released model checkpoints this metric builds on. We use an mDeBERTa-v3-base model (He et al, 2021) pretrained on the 100-language CC100 corpus and fine-tuned for natural language inference on XNLI (Conneau et al, 2018) (which includes Arabic) and English MNLI (Williams et al, 2018): the released Hugging Face checkpoint `MoritzLaurer/mDeBERTa-v3-base-mnli-xnli`. This choice matters: an English-only NLI classifier applied to Arabic or Malay premise–hypothesis pairs produces scores that reflect the classifier's inability to parse non-English text rather than a genuine faithfulness judgment, an issue we diagnosed during pipeline validation by observing that early Arabic F_{faith} scores clustered within 0.01 of the 1-in-3 random baseline for a three-way {entailment, neutral, contradiction} classifier, consistent with a classifier defaulting to near-uniform predictions on text it cannot interpret.

The training reward and the headline score use the same classifier and the same premise (the passage). A second, independent classifier (`joeddav/xlm-roberta-large-xnli`, a different architecture trained on different data) scores every held-out answer and is never used for training; we report it as Faith_2. Both scorers truncate the premise to 512 tokens.

An entailment classifier can reward answers that closely paraphrase or partially copy the context regardless of whether they address the question, and the same score is both the reward and the primary outcome. We therefore call it the *NLI faithfulness score*, report it beside F1, exact match, answer length and the independent scorer, and analyse it against answer length (Section 3.5). We make no claim about human-judged faithfulness for the two main backbones; Appendix G reports a blinded human validation of the first protocol's generations, and Section 4.7 the human judgments of the legacy mT5 adapter.

Correctness metrics. Exact match (EM) and token-level F1 between the generated and the gold answer follow standard SQuAD-style normalization (lowercasing, punctuation stripping, whitespace collapsing) and are computed independently of the NLI score. For Malay we also report *option accuracy*, the share of answers that select the gold option. EM and F1 reward short spans, so they penalize any answer that is longer than the reference even when it is correct.

3.5 Length-Matched Analysis

Entailment scores depend strongly on answer length: a one-word answer is rarely entailed by a passage under an NLI model, and a sentence copied from the passage is.

Because fine-tuning and, in the legacy study, the reward both change answer length, we analyse the NLI score at fixed length. Answers are assigned to five bins by their number of words (1, 2–3, 4–6, 7–10, 11+). We report (i) the mean NLI score by bin and condition; (ii) *matched pairs*: for two conditions, the answers to the same held-out item and seed that fall in the same bin, with the mean paired difference (reported over all matched pairs and per bin when at least 30 pairs exist); and (iii) a *length-adjusted* contrast, the coefficient of the condition indicator in a least-squares regression of the NLI score on the condition and bin fixed effects, estimated on the same pairs as the unadjusted contrast. All intervals are bootstrap intervals over held-out items (items resampled with all their seeds). The matched estimate answers "do the two conditions differ when they produce answers of similar length" for the subset of items where they do, and the adjusted estimate uses all pairs under a constant-bin-effect model. Neither is an intervention on length. The analysis was added after the main results were seen and is exploratory in its specification.

3.6 Statistical Analysis

The primary quantity is the paired difference between two conditions in a score, averaged over seeds for each held-out example. We report the mean with a 95% percentile bootstrap interval over held-out items (2,000 resamples), which describes sampling of the evaluation items given the trained models. For the NLI score we also report a seed-based 95% t interval (4 degrees of freedom) of the per-seed difference between the two conditions, because five seeds is the unit that describes variation of the training procedure. Contrasts are C minus B, C minus $C_{\lambda=0}$ and C minus C_{shuf}, per backbone and language. No multiplicity correction is applied to the 12 contrasts per outcome; intervals that exclude zero should be read with that in mind. We do not declare an equivalence margin for the main study. The design (backbones, conditions, seeds and training schedule) was fixed before the runs; the contrasts above and the length analysis were specified after the first results were seen. Frozen condition A has a single deterministic run and is compared descriptively and with item-level intervals only.

3.7 Protocol History

The study replaces two earlier ones. The first (188 runs on mT5-base, Appendix C) reported that the reward harmed LoRA and DoRA; controls and an audit of six implementation defects (Appendix A) showed that this was an evaluation artifact: a model left in training mode with gradient checkpointing enabled decodes from the last token only. The corrected mT5 study (162 launcher jobs, 228 records, eight seeds; Appendix B) is the legacy baseline used in Section 4.7. The present runner implements all corrections by construction: the raw passage is the premise, sampling and scoring share one distribution and one model state, decoding is in evaluation mode with an assertion, and every prediction is stored with its identifier.

3.8 AI-Assisted Tooling

The authors used AI-assisted tooling (a large language model) during preparation of this work, for three purposes: code implementation, including drafting and debugging

portions of the training, evaluation, and analysis scripts released with this paper; literature search and citation verification; and manuscript preparation, including copy-editing and reference formatting. Annotators were instructed not to use AI tools when labelling. No AI system is an author of this paper. All AI-assisted code, citations, and text were reviewed, verified, and edited by the human authors, who take full responsibility for the accuracy and integrity of the final manuscript, its code, and its reported results.

4 Results

All results in this section come from the primary study (Section 3): Qwen3-0.6B-Base and T5Gemma 2 270M-270M, Arabic and Malay, five seeds per trained condition, 84 runs. Table 2 gives the cell means, Table 3 the paired contrasts and Figure 1 the seed-level scores.

4.1 Run integrity

All 84 runs completed and wrote 720 (Arabic) or 430 (Malay) held-out predictions; the frozen T5Gemma 2 model returned an empty answer for 5.6% of the Arabic and 5.1% of the Malay items, and no trained run returned any. Training took 44.7 GPU-hours in total. Twelve Qwen3 runs were restarted after out-of-memory failures and the $C_{\lambda=0}$ run of four seed-language cells was executed twice (Section 3.3). For those four cells the two executions of the same seed give NLI scores that differ by 0.004, 0.003, 0.011 and 0.012 and F1 scores that differ by 0.014, 0.009, 0.021 and 0.007 (Arabic seeds 2026 and 99, Malay seeds 42 and 99). This run-to-run difference at a fixed seed is as large as the largest nominal reward effect reported below (+0.011), and it is a reason to treat differences of about 0.01 in the NLI score as at the limit of what these runs resolve.

4.2 Fine-tuning helps; the reward is not what helps

Fine-tuning improves every reference-based score by a large margin (Table 2). F1 rises from 0.21 to 0.59 for Qwen3 and from 0.04 to 0.56 for T5Gemma 2 in Arabic, and from 0.38 to 0.64 and from 0.11 to 0.48 in Malay; exact match rises from at most 0.075 to between 0.38 and 0.58 in every cell, and Malay option accuracy from 0.41 to 0.58 (Qwen3) and from 0.19 to 0.39 (T5Gemma 2). Frozen models produce long answers (13 words in Arabic, 7 to 18 in Malay) and fine-tuned ones three to four words, as the gold answers are.

The NLI score behaves differently in the two languages. In Arabic, fine-tuning *lowers* it (Qwen3 0.506 $\rightarrow$ 0.144, T5Gemma 2 0.186 $\rightarrow$ 0.132); in Malay it is unchanged for Qwen3 (0.292 $\rightarrow$ 0.293) and rises for T5Gemma 2 (0.092 $\rightarrow$ 0.295). Section 4.5 shows that most of this pattern is an answer-length effect. Among trained conditions, B, C, $C_{\lambda=0}$ and C_{shuf} are close in every cell: the largest difference in the NLI score between any two of them is 0.012 (Qwen3, Arabic, C against $C_{\lambda=0}$).

Table 2 Primary study: mean over seeds (and standard deviation over seeds where more than one) of held-out scores. Faith: NLI entailment score under the reward model; Faith_2: independent NLI model; Opt.: option accuracy (Malay only; the answer selects the gold option). Condition A has one seed because it involves no training. Held-out sets: 720 Arabic and 430 Malay items.

Condition	n_{seeds}	Faith	Faith_2	F1	EM	Opt.	Words
Qwen3-0.6B-Base, Arabic							
A (frozen)	1	0.506	0.439	0.211	0.075	–	12.8
B (CE)	5	0.144 ±0.003	0.169 ±0.007	0.592 ±0.015	0.422 ±0.014	–	3.1
C (CE+NLI)	5	0.156 ±0.002	0.172 ±0.003	0.604 ±0.009	0.432 ±0.010	–	3.3
C, λ=0	5	0.144 ±0.006	0.172 ±0.010	0.591 ±0.016	0.425 ±0.016	–	3.1
C, shuffled	5	0.147 ±0.004	0.170 ±0.003	0.599 ±0.011	0.435 ±0.010	–	3.1
Qwen3-0.6B-Base, Malay							
A (frozen)	1	0.292	0.319	0.384	0.005	0.409	7.1
B (CE)	5	0.293 ±0.004	0.352 ±0.008	0.643 ±0.014	0.578 ±0.015	0.579 ±0.015	4.2
C (CE+NLI)	5	0.302 ±0.007	0.360 ±0.011	0.622 ±0.029	0.557 ±0.035	0.559 ±0.034	4.1
C, λ=0	5	0.295 ±0.015	0.353 ±0.018	0.638 ±0.009	0.573 ±0.012	0.573 ±0.012	4.1
C, shuffled	5	0.299 ±0.009	0.356 ±0.006	0.641 ±0.016	0.579 ±0.021	0.580 ±0.021	4.1
T5Gemma 2, Arabic							
A (frozen)	1	0.186	0.206	0.036	0.000	–	13.0
B (CE)	5	0.132 ±0.010	0.157 ±0.008	0.563 ±0.015	0.382 ±0.013	–	2.9
C (CE+NLI)	5	0.131 ±0.017	0.151 ±0.014	0.566 ±0.019	0.378 ±0.011	–	2.8
C, λ=0	5	0.130 ±0.005	0.156 ±0.009	0.572 ±0.009	0.384 ±0.005	–	2.8
C, shuffled	5	0.136 ±0.004	0.162 ±0.012	0.572 ±0.009	0.381 ±0.009	–	3.0
T5Gemma 2, Malay							
A (frozen)	1	0.092	0.112	0.113	0.000	0.186	17.7
B (CE)	5	0.295 ±0.016	0.338 ±0.011	0.481 ±0.030	0.385 ±0.040	0.392 ±0.037	4.1
C (CE+NLI)	5	0.301 ±0.010	0.344 ±0.008	0.481 ±0.013	0.387 ±0.014	0.391 ±0.015	4.2
C, λ=0	5	0.291 ±0.010	0.344 ±0.014	0.472 ±0.021	0.373 ±0.019	0.383 ±0.023	4.1
C, shuffled	5	0.293 ±0.006	0.352 ±0.020	0.492 ±0.014	0.395 ±0.017	0.402 ±0.018	4.1

4.3 The reward adds nothing reliable beyond cross-entropy and the controls

Table 3 reports the contrasts of the reward objective C against cross-entropy training and against both controls. In the NLI score the point estimates range from −0.005 to +0.012. Nine of the twelve item-bootstrap intervals include zero; the three that do not (C−B +0.011 [+0.005, +0.018], C−$\text{C}_{\lambda=0}$ +0.012 [+0.005, +0.018] and C−C_{shuf} +0.008 [+0.002, +0.015]) are all in Qwen3 on Arabic, where C is above B in all five seeds. The seed-based intervals agree for that cell (C−B [+0.008, +0.015]) and are wider elsewhere, as high as ±0.03 for T5Gemma 2 on Arabic. Because both controls are as far below C as B is, the Qwen3 Arabic difference cannot be attributed to the machinery alone; whether it is attributable to the information in the reward is examined in Section 4.5. For T5Gemma 2 on Malay, C is above both controls in five of five seeds (+0.010 and +0.008), but the intervals include zero. The two controls are themselves indistinguishable from cross-entropy training: their differences from B in the NLI score lie between −0.004 and +0.006 in every cell, so running the sampling, scoring and policy-gradient path with the reward information removed does nothing measurable.

F1 and exact match do not improve with the reward either (Table J2 in Appendix J). C−B in F1 is +0.011 in Qwen3 Arabic, −0.021 [−0.040, −0.003] in

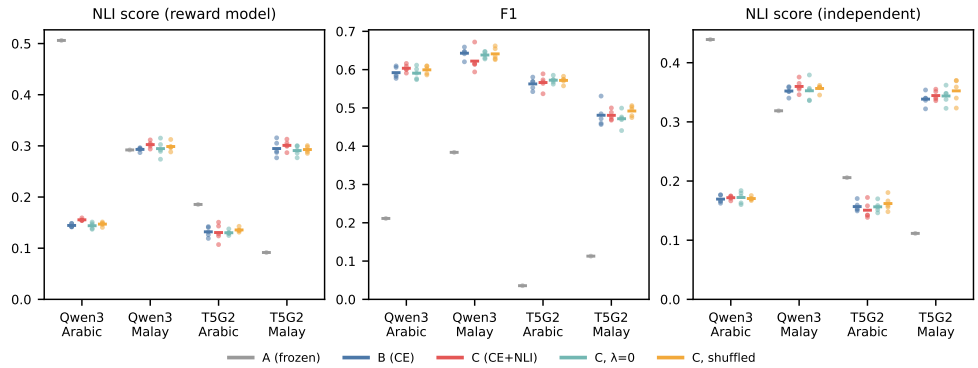


Fig. 1 Held-out scores by cell and condition. Each dot is one seed and the bar is the mean; the frozen condition has one run. Left: NLI score under the reward model; middle: F1; right: NLI score under the independent scorer. Conditions B, C and the two controls overlap within seed spread in every cell and on every measure.

Table 3 Paired contrasts of the NLI score (Ar: Arabic, Ma: Malay, T5G2: T5Gemma 2; mean difference). Faith: reward-model score, 95% bootstrap interval over held-out items (seed-averaged item scores); *: interval excludes zero. Faith, seeds: same contrast, 95% t interval over the five seeds (4 degrees of freedom); †: excludes zero. Faith$_2$: independent scorer, item bootstrap. None is adjusted for the 12 contrasts per outcome. C−B tests the composite objective as a whole; C against the two controls tests whether the reward information matters.

Cell	Contrast	Faith	Faith, seeds	Faith$_2$
Qwen3-Ar	C−B	+.011* [+.005,+.018]	[+.008,+.015]†	+.002 [−.004,+.009]
Qwen3-Ar	C−C$_{\lambda=0}$	+.012* [+.005,+.018]	[+.002,+.021]†	.000 [−.007,+.007]
Qwen3-Ar	C−C$_{\text{shuf}}$	+.008* [+.002,+.015]	[+.004,+.013]†	+.001 [−.005,+.008]
Qwen3-Ma	C−B	+.009 [−.003,+.022]	[−.003,+.022]	+.008 [−.004,+.020]
Qwen3-Ma	C−C$_{\lambda=0}$	+.008 [−.005,+.021]	[−.011,+.027]	+.007 [−.004,+.020]
Qwen3-Ma	C−C$_{\text{shuf}}$	+.004 [−.010,+.017]	[−.009,+.017]	+.003 [−.009,+.016]
T5G2-Ar	C−B	−.001 [−.008,+.005]	[−.031,+.028]	−.006 [−.013,+.001]
T5G2-Ar	C−C$_{\lambda=0}$	.000 [−.006,+.007]	[−.025,+.026]	−.006 [−.013,+.002]
T5G2-Ar	C−C$_{\text{shuf}}$	−.005 [−.011,+.001]	[−.027,+.017]	−.011* [−.019,−.004]
T5G2-Ma	C−B	+.006 [−.008,+.021]	[−.010,+.022]	+.006 [−.006,+.019]
T5G2-Ma	C−C$_{\lambda=0}$	+.010 [−.003,+.024]	[−.004,+.024]	+.001 [−.012,+.013]
T5G2-Ma	C−C$_{\text{shuf}}$	+.008 [−.005,+.023]	[−.002,+.019]	−.008 [−.020,+.005]

Qwen3 Malay, and +0.004 (Arabic) and 0.000 (Malay) for T5Gemma 2; the corresponding exact-match differences are +0.009, −0.021 [−0.043, 0.000], −0.004 and +0.001. The Qwen3 Malay decrease is small and not distinguishable from the two controls (−0.016 and −0.019 in F1), and in that cell the seed-to-seed standard deviation of F1 under C is 0.029 against 0.009 to 0.016 for the other conditions, so a small loss cannot be separated from seed variation with five seeds.

4.4 An independent NLI scorer does not reproduce the Qwen3 Arabic difference

The independent scorer (Faith_2) ranks the conditions the same way as the reward model for the large effects (the frozen versus trained gap) and not for the small ones. For Qwen3 on Arabic, C−B is +0.002 [−0.004, +0.009], against +0.011 under the reward model; the differences from $C_{\lambda=0}$ and C_{shuf} are 0.000 and +0.001. The only interval on this scorer that excludes zero is in the opposite direction: for T5Gemma 2 on Arabic, C is 0.011 [0.004, 0.019] *below* the shuffled-reward control. A gain that appears under the scorer used for training and not under a second scorer is the signature of a reward-specific effect on the score and not of better answers; here it is also very small, so the most economical reading is that the Qwen3 Arabic difference is a small, scorer-specific shift and not a robust improvement.

4.5 Controlling for answer length

Table 4 and Figure 2 analyse the NLI score at fixed answer length (Section 3.5). Two conclusions follow.

The remaining reward effect is a length effect.

In Qwen3 Arabic, C answers are slightly longer than B answers (3.34 against 3.12 words, Table 2), and most answers of both conditions fall in the same bin (77% of the pairs). Within matched pairs the C−B difference is +0.002 [−0.003, +0.007]; with length-bin fixed effects it is +0.003 [−0.004, +0.009], against +0.011 unadjusted. The same reduction holds against the controls (+0.002 and −0.001, adjusted). The Qwen3 Arabic gain is thus compatible with C producing marginally longer answers rather than more entailed answers of equal length. The adjusted C−B interval for Qwen3 on Malay excludes zero (+0.012 [+0.001, +0.026], unadjusted +0.009), but the adjusted differences from the two controls (+0.009 and +0.004) and the matched-pair estimate (+0.005) do not, and the adjustment moves the estimate by only 0.003. Apart from the Qwen3 Malay estimates just discussed, every adjusted or matched contrast for C lies between −0.005 and +0.006 and has an interval that includes zero.

Most of the NLI drop after fine-tuning is length.

In Qwen3 Arabic the frozen model's answers average 12.8 words and score 0.506; after fine-tuning they average 3.1 words and score 0.144. At fixed length the difference B−A shrinks from −0.362 [−0.397, −0.326] to −0.065 [−0.106, −0.023]. In T5Gemma 2 the frozen model is a poor answerer (EM 0, 13 to 18 words per answer, empty answers on some items) and the sign of the Arabic difference reverses once length is held fixed, from −0.054 to +0.101 [+0.070, +0.134]; in Malay it is +0.203 unadjusted and +0.221 adjusted. These contrasts rest on one frozen run and on pairs that rarely share a bin (4% to 43%), so we treat them as descriptive. They show that a lower NLI score after fine-tuning in Arabic is not evidence that fine-tuning made the model less faithful, and that comparisons of NLI scores between systems that give answers of different length (frozen against trained, or long-form against span-style) should be made at matched length. Table J3 in Appendix J gives the full length-bin means.

Table 4 Faithfulness contrasts with answer length controlled (Ar: Arabic, Ma: Malay, T5G2: T5Gemma 2). Unadjusted: mean paired difference. Adjusted: coefficient of the condition indicator in a regression of the NLI score on the condition and answer-length-bin fixed effects (bins 1, 2–3, 4–6, 7–10, 11+ words; the same items and seeds as the unadjusted contrast). Matched: mean difference over pairs of answers to the same item and seed that fall in the same length bin. Same bin: share of pairs that fall in the same bin. All intervals are 95% bootstrap intervals over held-out items. For B−A the frozen model has one seed, and its answers are rarely in the same bin as trained answers.

Cell	Contrast	Unadjusted	Length-adjusted	Matched	Same bin
Qwen3-Ar	C−B	+.011 [+.005,+.017]	+.003 [−.004,+.009]	+.002 [−.003,+.007]	77%
Qwen3-Ar	C−$C_{\lambda=0}$	+.012 [+.006,+.018]	+.002 [−.003,+.008]	+.004 [.000,+.009]	78%
Qwen3-Ar	C−C_{shuf}	+.008 [+.002,+.015]	−.001 [−.007,+.005]	+.004 [−.001,+.009]	78%
Qwen3-Ar	B−A	−.362 [−.397,−.326]	−.065 [−.106,−.023]	−.019 [−.057,+.019]	15%
Qwen3-Ma	C−B	+.009 [−.003,+.022]	+.012 [+.001,+.026]	+.005 [−.006,+.015]	87%
Qwen3-Ma	C−$C_{\lambda=0}$	+.008 [−.004,+.020]	+.009 [−.005,+.022]	+.002 [−.008,+.013]	87%
Qwen3-Ma	C−C_{shuf}	+.004 [−.009,+.018]	+.004 [−.011,+.019]	+.001 [−.009,+.011]	87%
Qwen3-Ma	B−A	+.001 [−.032,+.033]	+.077 [+.047,+.107]	+.033 [−.007,+.074]	43%
T5G2-Ar	C−B	−.001 [−.008,+.006]	+.002 [−.004,+.008]	−.001 [−.007,+.005]	72%
T5G2-Ar	C−$C_{\lambda=0}$	.000 [−.006,+.007]	+.001 [−.004,+.008]	−.003 [−.009,+.003]	73%
T5G2-Ar	C−C_{shuf}	−.005 [−.011,+.001]	+.002 [−.004,+.007]	−.005 [−.013,+.003]	71%
T5G2-Ar	B−A	−.054 [−.079,−.029]	+.101 [+.070,+.134]	+.094 [+.041,+.151]	9%
T5G2-Ma	C−B	+.006 [−.008,+.022]	+.003 [−.011,+.017]	−.003 [−.014,+.008]	80%
T5G2-Ma	C−$C_{\lambda=0}$	+.010 [−.002,+.024]	+.006 [−.007,+.020]	−.002 [−.013,+.009]	81%
T5G2-Ma	C−C_{shuf}	+.008 [−.005,+.022]	+.006 [−.007,+.020]	−.002 [−.013,+.010]	80%
T5G2-Ma	B−A	+.203 [+.169,+.240]	+.221 [+.161,+.286]	+.345 [+.185,+.512]	4%

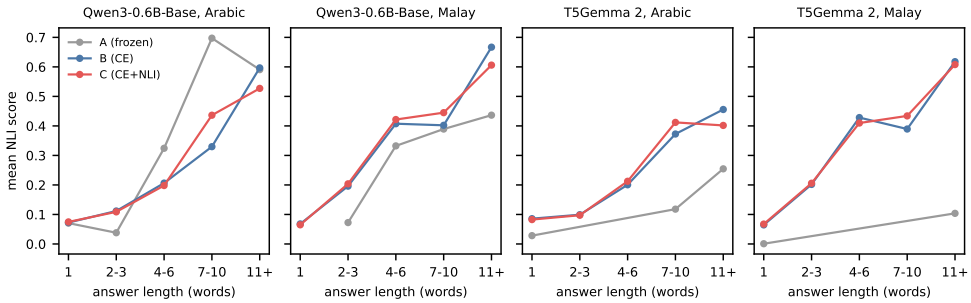


Fig. 2 Mean NLI score by answer length (words) for the frozen model (A), cross-entropy training (B) and the reward objective (C), pooled over seeds; bins with fewer than 30 answers are omitted. Within a length bin, B and C overlap in every cell; A differs from them, but its answers are rarely in the same bin (Table 4).

4.6 Why the reward is inert: training diagnostics

Table 5 summarizes the training logs of condition C. In the first quarter of training the policy-gradient term is non-negligible (mean |PG| between 0.19 and 0.56), and cross-entropy is still falling. By the second half of training cross-entropy on the minibatch

Table 5 Training diagnostics of Config C (logged every 50 steps; all seeds; quarters of the 1,416 steps). CE: token-level cross-entropy of the minibatch; |PG|: absolute policy-gradient term; Reward: mean NLI reward of the sampled answers; Inactive: share of logged steps with $|\mathrm{PG}| < 10^{-3}$, i.e. all K=4 samples receive (almost) the same reward or the same answer.

Model	Lang.	Qtr	CE	|PG|	Reward	Inactive
Qwen3-0.6B-Base	Arabic	1	0.266	0.527	0.199	9%
		2	0.060	0.080	0.154	43%
		3	0.017	0.017	0.175	80%
		4	0.008	0.012	0.178	88%
	Malay	1	0.104	0.192	0.247	34%
		2	0.017	0.020	0.314	83%
		3	0.006	0.017	0.292	86%
		4	0.003	0.005	0.280	90%
T5Gemma 2	Arabic	1	0.460	0.559	0.198	6%
		2	0.129	0.049	0.136	49%
		3	0.031	0.003	0.163	94%
		4	0.005	0.000	0.164	95%
	Malay	1	0.253	0.291	0.254	17%
		2	0.028	0.021	0.319	71%
		3	0.005	0.022	0.283	89%
		4	0.000	0.000	0.262	100%

is at or below 0.03 in every cell and, for Qwen3 and T5Gemma 2 alike, the policy-gradient term is below 10^{-3} in 80% to 100% of the logged steps. With leave-one-out advantages the policy term vanishes when the K=4 sampled answers receive the same reward, which is what happens when the model has memorized the training answer and reproduces it in all four samples. The mean reward of the sampled answers changes little over training (first to last quarter: 0.20 → 0.18 for Qwen3 on Arabic, 0.25 → 0.28 on Malay, 0.20 → 0.16 for T5Gemma 2 on Arabic, 0.25 → 0.26 on Malay), which fits a reward signal that is mostly absent and not one that is exploited. In the cross-entropy runs the training loss in the last quarter of training is 0.002 (Qwen3, Arabic), 0.00004 (Qwen3, Malay), 0.003 (T5Gemma 2, Arabic) and 2.5×10^{-6} (T5Gemma 2, Malay). The logs record the minibatch loss every 50 steps and not the within-prompt reward variance, so this is indirect evidence: we did not measure the variance directly, and a different schedule (fewer epochs, a lower learning rate, more training data or a larger λ_1) could keep the policy term alive. We did not test these.

4.7 Contrast with the legacy mT5-base study

The corrected mT5-base study of the same objective (eight seeds per cell, three adapters; Appendix B) gave a different result in Arabic. The NLI score rose from about 0.15 to 0.72–0.77 (+0.60, 95% CI +0.56 to +0.64); the zero-weight and shuffled-reward controls stayed within 0.01 of cross-entropy training, so the effect was specific to the reward; and an independent scorer agreed (+0.257). But exact match fell from 0.07 to 0.004, answers grew from 2.9 to 16.5 words and the share of answers occurring verbatim in the passage fell from 87% to 12%; adjusting for answer-length bins reduced the estimated gain by about three quarters (+0.60 → +0.14); and blinded

human raters (60 items per language, two annotators, $\kappa = 0.38$) judged the answers no more correct and, on average, less supported (Appendix B). In Malay the mT5 effect was small (+0.008, 95% CI −0.001 to +0.016). The mT5 result is therefore an over-optimization of the NLI score: a large, reward-specific gain in the score and no gain in the other measures.

What differs between the two sets of backbones in our logs is whether the policy memorizes its training set. The mT5 adapters were fitted less tightly: in the legacy trainer's logs the cross-entropy loss over the last epoch averages 1.67 (Arabic) and 0.74 (Malay) for condition B, against 0.002 and 0.00004 for Qwen3 in the last quarter of training (the two logging implementations differ and the comparison is descriptive), and mT5 reaches an Arabic exact match of only 0.07 against 0.38 to 0.43 for the new backbones. A policy that remains stochastic keeps non-zero advantages and can move towards answers the scorer likes; one that has memorized the training set cannot. We present this as a hypothesis that fits the available logs, not as a tested mechanism: the two studies differ in backbone, tokenizer, trainer, adapters and checkpointing history, and no experiment here changes only the degree of memorization.

4.8 Summary of evidence

The primary study supports four statements. First, LoRA fine-tuning on 470 examples produces large gains in F1, exact match and option accuracy over the frozen models for both backbones and both languages. Second, the NLI-entailment reward adds no reliable improvement to the NLI faithfulness score, to F1 or to exact match beyond cross-entropy training and two controls that remove the reward's information; the single nominally significant gain (Qwen3, Arabic, +0.011) is not reproduced by an independent scorer and disappears when answer length is held fixed. Third, most of the change in the NLI score between frozen and fine-tuned models is a change in answer length. Fourth, the training logs show that the policy-gradient term is inactive for most of the second half of training, whereas the legacy mT5 adapters, which do not memorize the training set, show a large reward effect that is an over-optimization of the score. The study has limits: one reward weight, one training schedule, one reward model, small backbones, and no human evaluation of the new backbones (Section 6).

5 Discussion

An NLI reward added to LoRA fine-tuning did not help in this regime. On two modern backbones, in two languages, with five seeds, the reward changed the NLI faithfulness score by at most 0.012, and the best-supported contrast (Qwen3, Arabic) was not reproduced by an independent scorer and disappeared at matched answer length. The seed-based upper limits of the C−B intervals are 0.015, 0.022, 0.028 and 0.022 in the four cells, so improvements larger than about 0.03 in the NLI score are excluded by these data, against the +0.60 seen with mT5. We read this as a null result under the tested schedule and not as proof that an NLI reward cannot help: the test used one reward weight ($\lambda_1 = 0.3$), one reward model, one 12-epoch schedule and a 470-example training set.

A reward needs a policy that still varies. The training logs suggest why the reward was inert. Cross-entropy training for 12 epochs on 470 examples drives the loss to nearly zero, the sampled answers reproduce the memorized training answer, and the leave-one-out policy-gradient term is almost always zero in the second half of training. A reward-based term in a composite objective cannot act where the policy has no variance. The mT5 baseline, which was fitted less tightly, kept a stochastic policy and showed a large reward effect. This is an observation about logs, not a controlled test, but it suggests a diagnostic that costs nothing: log the within-prompt reward variance, or the share of steps in which the policy-gradient term is exactly zero, and report it. A reward effect that is absent should not be reported as a property of the reward until that share is known.

The reward, the metric and answer length are entangled. When an entailment score is both the reward and the outcome, and when it depends strongly on answer length, any system that changes answer length changes the score. The length-matched analysis shows two consequences. In Qwen3 on Arabic, the one nominal reward effect is a difference of about 0.2 words in mean answer length. And the apparent loss of faithfulness after fine-tuning in Arabic ($0.506 \rightarrow 0.144$) is, to a large part, a change from 13-word generations to three-word spans: at fixed length the drop is 0.065 and, for T5Gemma 2, the sign reverses. Faithfulness scores of systems that answer at different lengths should therefore be compared within length bins or with length as a covariate, and a reward that is meant to improve faithfulness should be paired with a length-controlled evaluation.

Where the reward works, it over-optimizes. The mT5 baseline is the opposite case: the reward raised the NLI score by +0.60 in Arabic, but exact match fell to near zero, the answers grew sixfold, and adjusting for length bins removed about three quarters of the estimate. Blinded human raters judged those answers no more correct and, on average, less supported. This is the pattern expected if the policy over-optimizes a proxy, as in the reward-model over-optimization literature (Gao et al, 2023a). The human evidence is limited (one adapter, one seed, 60 items per language, two raters who disagree) and it is not a demonstration that the answers are less faithful. The two sets of results together say that an NLI reward can do nothing or can do harm, and that the NLI score alone cannot tell the two apart.

Resolution limits. The same seed and condition, trained twice, differed by up to 0.012 in the NLI score and 0.021 in F1 (Section 4.1). Differences of about 0.01 in these scores are at the limit of what a single run resolves, which is why we report five seeds and interpret contrasts of that size cautiously.

Evaluation state is part of the experiment. The first study's central result, a harm of 0.16 from the reward, was an artifact of decoding a model left in training mode with gradient checkpointing enabled (Appendix C). The primary study asserts evaluation mode and no checkpointing at decode time, and no reported run used checkpointing. We recommend asserting the mode in every evaluation loop, recording memory-saving flags in every run record, and running the controls that remove the reward's information.

Practical guidance. For groups adding a sample-based reward to a fine-tuning run: (i) include a zero-weight and a shuffled-reward control that execute the same

code path; (ii) log the share of steps in which the reward term has a non-zero gradient; (iii) score the same premise in training and in evaluation and use a second scorer for evaluation; (iv) report answer length and analyse the score at matched length; (v) report reward-independent metrics beside the reward; (vi) assert evaluation mode before decoding and treat memory-saving switches as part of the experimental condition; (vii) repeat at least one run to measure run-to-run noise; and (viii) validate any automatic faithfulness score against human judgment in the target language before reading differences as improvements.

What would settle the open questions. A direct test of the memorization account would train the same backbones with fewer epochs, a lower learning rate or more data, and log the reward variance; a test of the length account would use a length-penalized or length-capped reward; a test of generality would add larger and instruction-tuned backbones and a reward-weight grid. These need new GPU runs and are not part of this paper.

6 Limitations

- **One reward, one weight, one schedule.** The primary study uses a single NLI reward model, $\lambda_1 = 0.3$, $K = 4$, one learning rate, 12 epochs and 470 training examples. The null result applies to that regime. We did not vary the schedule, the reward weight or the amount of training data, so the memorization account of the null is supported by training logs and a comparison with a different study, not by an experiment that changes memorization alone.
- **Small backbones.** Qwen3-0.6B-Base and T5Gemma 2 270M-270M are small. The result may not transfer to larger or instruction-tuned multilingual models.
- **The metric is the reward.** The NLI faithfulness score is the training reward and the headline outcome. A second NLI model scores the answers independently, but both are entailment classifiers, and neither was trained on Malay NLI data. We did not evaluate the new backbones with human raters, so the study supports no claim about human-judged faithfulness for them.
- **The legacy comparison is confounded.** The mT5 baseline differs from the primary study in backbone, tokenizer, trainer, adapters, number of seeds (eight against five), checkpointing history and logging implementation. The contrast in training behaviour (Section 4.7) is descriptive.
- **Length analysis is descriptive and exploratory.** The length-matched and length-adjusted estimates were specified after the main results were seen. They do not isolate a causal route, because the conditions produce answers of equal length only for a subset of items (4% to 87% of the pairs), and the frozen condition has one run.
- **Multiplicity and resolution.** The 12 contrasts per outcome are not corrected for multiplicity. Five seeds give seed-based intervals as wide as ± 0.03 for T5Gemma 2 on Arabic, and a repeated run with the same seed can differ by 0.012 in the NLI score.
- **Run mishaps.** Twelve runs were restarted after out-of-memory failures, and four $C_{\lambda=0}$ runs were executed twice with the later result stored (Section 3.3). Neither

affects the reported contrasts qualitatively, since the two executions agree to within about 0.01 in the NLI score, but the first-versus-second execution is not recorded.

- **Task and data.** Closed-passage QA with gold passages, not full RAG. Arabic (extractive, translated XQuAD) and Malay (Belebele with options in the prompt) differ in format, so language differences are benchmark-language differences. The index-based split shares one Arabic passage across the boundary (four held-out items, 0.6%); excluding them changes the NLI means by at most 0.001. About 2–5% of prompts exceed 512 tokens and have the passage truncated.
- **Legacy human evidence.** The human validation (Appendix G and Section 4.7) covers mT5 generations only, with two volunteer annotators, low agreement ($\kappa =$ 0.38 for support, 0.18 for correctness) and no rater-proficiency screening. The ethics status of the annotation tasks is given in the Statements and Declarations.

7 Reproducibility

From saved results (CPU, minutes). The release contains, for every one of the 84 primary runs, a run record, the training log and every held-out prediction with its example identifier, the faithfulness score under both scorers, F1, exact match and answer length. `analyze.py` regenerates the cell means, the contrasts and the bootstrap intervals; `lenmatch.py` regenerates the length-matched analysis (Table 4, Table J3); and `gen_paper_assets.py` regenerates every table and figure of Section 4 from those files. The legacy mT5 study's 228 run records, its prediction files and the first study's 188 records are also released, with the scripts that regenerate the appendices.

From training (GPU). `run_test.sh` runs a four-stage pre-flight (environment and gated-model access, smoke test of every condition, a two-epoch learning check, and a timing check), and `run_train.sh` launches the 84 runs over the available GPUs; completed runs are skipped on restart. The primary runs used PyTorch 2.12.1, Transformers 5.5.0 and PEFT 0.21.2 on NVIDIA H200 NVL GPUs; the legacy mT5 runs used the same stack. On other hardware or library versions we expect the same qualitative pattern and not the same decimals; the same seed run twice on the same stack already differs by up to 0.012 in the NLI score.

Code and data. The code, the exact index lists of the 470-example fine-tuning split and the held-out sets, and the per-run records are released in the anonymized code-and-data package supplied with the submission as supplementary material (a public repository link will be added after review). XQuAD (Apache 2.0) and Belebele (CC-BY-SA 4.0) are public and are not redistributed. The Qwen3 and T5Gemma 2 weights and the two NLI models are used from their public releases.

Prompts and generation. Prompts are in Section 3.1. Evaluation uses greedy decoding with at most 40 new tokens, stopping at the end-of-sequence token or at the first token containing a newline, in evaluation mode with the cache on. Policy sampling uses K=4, temperature 1.0, `top_p=1.0` and `top_k=0`, with the same stop rule.

AI-assisted tooling. Disclosed in Section 3.8.

8 Broader Impact

The results argue for caution. Fine-tuning small models on a few hundred examples improves Arabic and Malay question answering substantially, but an automatic faithfulness reward neither added to that improvement nor, on the new backbones, was shown to do harm; on the legacy mT5 baseline the same reward turned three-word answers into sixteen-word sentences that human raters judged no more correct and, on average, less supported by the passage. A metric that is also the training signal can certify a system that has learned to satisfy the metric, and automatic faithfulness scores should not replace human evaluation before deployment, especially in languages where the NLI model itself is weaker. Fluent but unsupported Arabic or Malay text is hard for non-expert users to detect, echoing findings that trust in AI output is miscalibrated by surface fluency (Zhang et al, 2020b). Negative and corrected results are also easy to lose: the first version of this work would have led practitioners to avoid an adapter to escape a "harm" that was an infrastructure setting, which is why we publish the correction and the null result with their controls.

For practitioners building Arabic and Malay RAG systems, the lessons are procedural: run controls that remove the reward's information, check that the reward term is active, report answer length and reward-independent metrics, record the environment per run, and validate any automatic faithfulness metric against human judgments in the target language. This complements retrieval-side robustness work: Arabic RAG benchmarks show retrieval noise and negative rejection remain open problems independent of the generator (Almohaimeed et al, 2026), long-context evidence aggregation targets the retrieval-generation interface (Hu et al, 2026), and Arabic question answering under knowledge updates has its own adaptation challenges (Ali et al, 2025). This paper addresses the generator only (Section 6).

9 Conclusion

We tested whether an NLI-entailment reward improves LoRA fine-tuning of Qwen3-0.6B-Base and T5Gemma 2 270M-270M on Arabic and Malay closed-passage question answering, with five seeds, a frozen reference and controls that remove the reward's information. Fine-tuning improved F1 and exact match sharply in every cell. The reward did not improve anything beyond it: its effect on the NLI faithfulness score was at most 0.012, the only nominally significant gain (Qwen3, Arabic) was not reproduced by an independent NLI scorer and disappeared when answer length was held fixed, and F1 and exact match did not improve. Most of the change in the NLI score between frozen and fine-tuned models was a change in answer length. Training logs show that the policy-gradient term is inactive for most of the second half of training, consistent with memorization of the 470 training examples. The legacy mT5-base baseline, which does not memorize, shows the opposite: a large reward-specific gain in the NLI score of +0.60 in Arabic, accompanied by sixfold longer answers, lower exact match and no gain in human-judged correctness, which we read as over-optimization of the score.

The broader lesson is that a reward effect, or its absence, should be reported against zero-weight and shuffled-reward controls, at matched answer length, with reward-independent metrics and with evidence that the reward term is active during training.

We release the corrected runner, per-example predictions for all 84 runs and the analysis code. A direct test of the memorization account, a length-penalized reward, larger and instruction-tuned backbones, a reward-weight grid and a human evaluation of the new backbones remain for future work.

Appendix A Legacy Baseline: mT5-base Protocol

This appendix and Appendix B describe the earlier mT5-base study of the same objective, which the main text uses as a legacy baseline (Section 4.7). Terminology in Appendices A to C follows the original wording: "the first study" is the 188-run study on the pre-correction path (Appendix C), and "the corrected study" is the mT5-base study of 162 launcher jobs (228 run records, eight seeds per cell) after the audit below; it is not the primary study of the main text. The dataset, prompts, faithfulness metric and splits are those of Section 3, with the exceptions noted here: mT5 used a sentinel-token suppression, an evaluation set of 716 Arabic items (the four items sharing a passage with the training slice were excluded) and the NLI premise handling described in the audit.

A.1 Backbone, Adapters and Configurations

mT5-base (Xue et al, 2021) (580M parameters) is the legacy backbone. Config A (frozen) was defined but not run; Config B is a LoRA adapter trained with cross-entropy; Config C uses the composite objective of Equation (1) with the leave-one-out REINFORCE estimator of Equation (D2) (Appendix D); Config D (full fine-tuning) is a single-seed reference from the first protocol (Appendix H). The corrected study runs Configs B and C for three adapters: plain LoRA, QLoRA (Dettmers et al, 2023) and DoRA (Liu et al, 2024), with rank r=8, α=16 and dropout 0.05, applied to the query, key, value and output projections and to the gated feed-forward block (`wi_0`, `wi_1`, `wo`). mT5's sentinel tokens (`<extra_id_N>`) are suppressed at sampling and at evaluation time via `suppress_tokens`. `lm_head` and the embeddings are not adapted. Pipeline corrections made before the first experiments are documented in Appendix E. AdaLoRA (Zhang et al, 2023) and VeRA (Kopiczko et al, 2024) were part of the first study only; both scored at the floor under cross-entropy training (EM = 0.000; Table C8), so a reward cannot be shown to change them, and they were not run in the corrected study.

A.2 Training Procedure and Controls (mT5-base)

All trainable configurations are fine-tuned for 12 epochs with batch size 4 (1416 optimizer steps) using AdamW at 2×10^{-4}, in bfloat16 mixed precision (T5-family models produced NaN gradients under float16 in our tests). The corrected study uses eight seeds (42, 123, 2026, 7, 11, 13, 17, 19); the first study used the first three. No validation split is held out, no checkpoint is selected, and no hyperparameter is tuned on the held-out evaluation set: every number is from the model state at the end of the fixed 12-epoch run.

Estimator. Config C samples K=4 answers per example with temperature 1.0, *no* nucleus or top-k truncation (`top_p=1.0`, `top_k=0`) and sentinel tokens suppressed; scores each with the frozen NLI reward; forms leave-one-out advantages; and re-scores the samples in a teacher-forced pass that (i) feeds `sampled[:, :-1]` as decoder input and scores `sampled[:, 1:]` (so the decoder-start token appears once), (ii) sanitizes logits and sets suppressed-token logits to a large negative value so that the scored distribution equals the sampled one, and (iii) masks only padding, counting the end-of-sequence token. A fail-fast assertion checks that every sampled sequence begins with the decoder-start token. Appendix D gives the derivation and the unit tests. The corrected study (Appendix B) uses this estimator with the further corrections of Appendix A.3: the raw passage as the NLI premise, and sampling and re-scoring with dropout off.

Gradient checkpointing. In the first protocol, Config C used activation (gradient) checkpointing for every adapter except QLoRA, to fit the K=4 samples in memory, and Configs B and QLoRA did not. We treated this as a memory setting that should not alter the optimization. It was not neutral in that study (Appendix C). In the corrected study the setting is an explicit flag stored in every run record; the core and control runs use checkpointing off, and a separate stage (Appendix A.4) trains B and C with it on.

Controls. Besides Config B, we run Config C with (i) λ_1=0, which executes the sampling, scoring and teacher-forced path with zero weight on the policy-gradient term; (ii) a *shuffled* reward, in which the $K\times$batch NLI rewards are permuted across the batch before advantages are formed, preserving their distribution while removing their dependence on the sampled answer; and, in a separate stage, (iii) checkpointing on and off. Any reward effect must exceed what the λ_1=0 and shuffled controls produce.

Run inventory and environment. The corrected study consists of 162 launcher jobs that wrote 228 run records, none failed (the unit of a job is one adapter-language-seed cell, or one control cell, and a job that trains both Config B and Config C writes two records). The core stage has 48 jobs (LoRA, QLoRA and DoRA $\times$ Arabic and Malay $\times$ eight seeds), each training Configs B and C, hence 96 records; the controls stage has 96 jobs (λ_1=0 and shuffled reward on the same 24 cells, eight seeds), each writing one Config C record, hence 96 records; and the checkpointing stage has 18 jobs (three adapters, two languages, seeds 42, 123 and 2026), each training B and C with checkpointing on, hence 36 records. In total $96 + 96 + 36 = 228$ records from 48+96+18 = 162 jobs; the released per-run records and prediction files were counted to confirm this inventory. The first study comprised 188 runs, described in Appendix C. Every run used one NVIDIA H200 NVL GPU with PyTorch 2.12.1, Transformers 5.5.0 and PEFT 0.21.2; each run record stores the GPU, driver and library versions, the dropout and checkpointing settings, and the evaluation state. The NLI scorer ran on the GPU. Hardware and software are therefore constant across all reported contrasts.

A.3 Implementation Audit and Status of the Reported Runs

A line-by-line audit of the training and evaluation code after the 188 runs of the first study found the defects in Table A1. They were found by static inspection and then

Table A1 Defects of the first (pre-correction) path, the runs they touch, and the corrected behavior (`LORA_V2=1`). Each correction has a CPU test (`tests_v2.py`) that fails on the old behavior and passes on the new.

#	Pre-correction behavior	Runs touched	Corrected behavior
1	Held-out generation ran in training mode (no `eval()`); with checkpointing enabled, T5 `generate()` decodes from the last token only	Every run (dropout on); collapse in checkpointed Config C	`eval()` and mode assertions before evaluation; the training-mode state is re-scored on the same adapter as a diagnostic
2	Training scored the decoded prompt (question, Malay options and passage) as the NLI premise; evaluation scored the passage	All Config C runs; reward obtainable from option text	The raw passage is carried through the dataset and collator and is the premise in both
3	Policy sampling and re-scoring drew independent dropout masks (estimator checked only in evaluation mode)	All Config C runs	Sampling and re-scoring run with dropout off; training mode is restored for the cross-entropy pass
4	Different non-finite-logit fills in generation ($\pm 10^4$) and scoring ($\pm 10^9$); no counts	All Config C runs	One shared sanitizer; per-run counters
5	AdaLoRA's rank scheduler was never called	AdaLoRA (fixed rank 12)	Scheduler called after each optimizer step; rank trajectory logged
6	Four Arabic held-out items share a passage with training; no stable item identifiers; checkpointing flag not stored	Arabic means; run records	Overlapping items excluded; source identifiers, passage hashes and flags stored

confirmed with CPU tests on a tiny T5 or on pretrained mT5-base. The corrected code is selected by `LORA_V2=1`; with it unset, the code reproduces the first study's behavior. The first study's adapter weights were not saved, so the effect of each defect on its results could not be measured directly. The corrected study (Appendix B) is the basis for every claim about the reward in this paper; the first study is retained in Appendix C as exploratory.

Three consequences follow. First, defect 1 explains the checkpointing effect (Appendix B.5). Second, defects 2–4 affect the training objective of every Config C run of the first study, so its checkpointing-off estimates describe a different objective from the one stated in Equation (D2): the reward is computed against the full prompt, so for Malay a sample could raise its reward by repeating option text, and the policy gradient is estimated from sequences that are not exactly on-policy. Third, defect 6 is handled for the first-study runs by re-scoring: their per-example outputs were retained, so we report all headline means without the four overlapping Arabic items (716 held-out Arabic examples) and release the complete per-example predictions with stable identifiers.

A.4 Design of the Corrected Study

The corrected study fixes its analysis in advance (Appendix A.5). It uses eight seeds per cell (42, 123, 2026, 7, 11, 13, 17, 19) for Configs B and C in LoRA, QLoRA and DoRA on both languages (48 jobs, 96 records), the λ_1=0 and shuffled-reward controls on the same cells (96 jobs, 96 records), and a checkpointing check of B and C with checkpointing on for three seeds (18 jobs, 36 records). Every adapter is additionally re-scored in the first study's evaluation state (model in training mode) and the score is stored beside the headline score. The protocol also listed three further stages: the

legacy dropout-on policy as an ablation, AdaLoRA and VeRA with an active rank schedule, and frozen and fully fine-tuned references. They were not run, and no claim in this paper depends on them.

A.5 Statistical Analysis (mT5-base)

Estimates and intervals. The unit of inference is the training seed; evaluation sets are fixed, so inference does not propagate evaluation-set sampling error. The primary quantity is the pooled difference in F_{faith} between two conditions: for each seed we average the paired differences over the six adapter-language cells (LoRA, QLoRA, DoRA $\times$ Arabic, Malay), and we report the mean with a 95% t interval on the seed-level values ($n-1$ degrees of freedom; 7 for the corrected study, 2 for the first study). Per-adapter intervals average the two language differences within each seed. p-values are secondary; the three per-adapter tests of a contrast are Holm-corrected. Counts of positive cells are descriptive only, because the 18 cells are not independent. Pooling treats adapters as exchangeable; we report the per-adapter estimates beside it. A secondary interval for the primary contrast comes from a bootstrap that resamples source items once per language and applies the same resample to all three adapters (seeds averaged within item), so the dependence of the adapters on shared items is preserved; a variant resamples whole passages, because several questions share a passage. These item-based intervals describe sampling of the evaluation items given the trained models and are kept separate from the seed-based primary interval.

Contrasts. (i) *C vs. B*: the composite objective minus cross-entropy training, the primary contrast, pooled over the six adapter-language cells and reported per language and per adapter. (ii) *Controls*: Config C minus the $\lambda_1{=}0$ and shuffled-reward controls, and each control minus B. (iii) *Checkpointing*: B and C with checkpointing on minus the same runs with it off.

Equivalence. The corrected study declares a margin of ± 0.01 in advance and calls a contrast equivalent only if its 90% interval lies inside it. (An earlier version of this analysis used ± 0.03, which is larger than the effects it was meant to bound; no equivalence claim is made from the first study.)

Status of the analysis. The corrected study's design, seed list, primary contrast, secondary outcomes and decision rules are fixed in the released runbook, and no seed was added or removed after data were seen. The first study's contrasts were specified after its results were known and are exploratory. Effects are reported in raw F_{faith} units. All quantities are computed from the per-run records by the released scripts (Section 7).

Appendix B Legacy Baseline: mT5-base Results

All results in this appendix come from the corrected mT5-base study (Appendix A.4): LoRA, QLoRA and DoRA, Arabic and Malay, eight seeds per cell, Configs B and C, the $\lambda_1{=}0$ and shuffled-reward controls, and a gradient-checkpointing check on three seeds. Results of the first, pre-correction study are in Appendix C; they are not pooled with the results below.

B.1 Run integrity

The corrected study has 162 launcher jobs and 228 run records (96 core, 96 control and 36 checkpointing records; Appendix A.3), and none failed. No run produced a non-finite logit in 5.8 million generation calls and 0.23 million scoring calls. A mean of 2.3% of held-out prompts (at most 4.6%) and 2.2% of training prompts (at most 4.5%) exceed the 512-token input limit and are truncated. Four Arabic held-out items share a passage with the training slice and are excluded from every headline mean (716 Arabic and 430 Malay held-out items).

B.2 The reward raises the NLI score in Arabic and not in Malay

Table B2 gives the main contrast. In Arabic, the composite objective raises F_{faith} from about 0.15 to 0.72–0.77 for all three adapters: +0.617 (QLoRA), +0.569 (LoRA) and +0.617 (DoRA), each with a 95% interval that excludes zero by a wide margin (all Holm-adjusted $p < 0.001$). Averaged over adapters the Arabic difference is +0.601 (95% CI $[+0.561, +0.641]$), and the pooled six-cell difference is +0.304 ($[+0.287, +0.322]$ over seeds; $[+0.290, +0.318]$ when items are resampled jointly across the three adapters and $[+0.277, +0.329]$ when whole passages are resampled), positive in all eight seeds. In Malay the difference is small: +0.013 for QLoRA ($[+0.003, +0.023]$, $p = 0.021$, Holm-adjusted $p = 0.063$), +0.004 for LoRA and +0.006 for DoRA, with intervals that include zero; averaged over adapters it is +0.008 ($[-0.001, +0.016]$).

The independent outcomes move the other way. In Arabic, exact match falls from about 0.07–0.08 to 0.003–0.006 (−0.070, −0.066 and −0.075) and F1 falls by 0.036 to 0.045; the pooled six-cell differences are −0.043 in exact match ($[-0.049, -0.037]$) and −0.024 in F1 ($[-0.029, -0.019]$). In Malay, exact match and F1 are unchanged for QLoRA (+0.003, +0.001) and lower for LoRA and DoRA (−0.027 and −0.023 in exact match; only DoRA's interval excludes zero, $[-0.045, -0.001]$, and it does not survive correction for three adapters). Averaged over the three adapters, Malay exact match is lower by 0.016 (seed-based 95% CI $[-0.029, -0.002]$) and F1 by 0.009 ($[-0.019, +0.001]$); these language-level contrasts are exploratory, uncorrected for the four language-by-metric comparisons, and the exact-match interval is close to zero, so we report a small decline in Malay exact match and do not claim that it is robust. The reward therefore raises the score it optimizes while the reference-based scores do not improve and, in Arabic, fall sharply. Because exact match and F1 reward short spans, they cannot by themselves show that the longer answers are wrong; Appendixes B.4 and B.7 examine this.

B.3 The effect is specific to the reward

Table B3 shows the controls. With the policy-gradient weight set to zero, or with the rewards shuffled across the batch, the NLI score stays at the cross-entropy level in Arabic (−0.003 and −0.002 against B; both intervals include zero) and moves by less than 0.01 in Malay. Pooled over the six cells, λ_1=0 and shuffled reward differ from B by −0.005 in F_{faith} (90% intervals $[-0.007, -0.002]$ for both), which lies inside the

Table B2 Main results of the corrected study (mT5-base, eight seeds per cell; mean ± SD over seeds). B: cross-entropy LoRA; C: composite (NLI-reward) objective. Δ: C minus B with a 95% t interval over the eight seed-paired differences (7 degrees of freedom), per language and adapter. Held-out examples: 716 Arabic and 430 Malay. F_{faith} is the training reward, so it is not an independent outcome; EM and F1 are.

Lang	Adapter	Metric	B	C	Δ (C−B) [95% CI]
Arabic	QLoRA	F_{faith}	0.149 ± 0.004	0.766 ± 0.084	+0.617 [+0.548, +0.686]
		EM	0.074 ± 0.004	0.003 ± 0.004	-0.070 [-0.073, -0.068]
		F1	0.124 ± 0.006	0.079 ± 0.005	-0.045 [-0.051, -0.039]
Arabic	LoRA	F_{faith}	0.151 ± 0.006	0.720 ± 0.105	+0.569 [+0.485, +0.653]
		EM	0.072 ± 0.010	0.006 ± 0.007	-0.066 [-0.077, -0.055]
		F1	0.126 ± 0.018	0.091 ± 0.015	-0.036 [-0.060, -0.012]
Arabic	DoRA	F_{faith}	0.149 ± 0.006	0.765 ± 0.073	+0.617 [+0.555, +0.679]
		EM	0.078 ± 0.017	0.003 ± 0.006	-0.075 [-0.091, -0.059]
		F1	0.135 ± 0.031	0.097 ± 0.013	-0.038 [-0.069, -0.007]
Malay	QLoRA	F_{faith}	0.318 ± 0.006	0.331 ± 0.012	+0.013 [+0.003, +0.023]
		EM	0.241 ± 0.016	0.243 ± 0.021	+0.003 [-0.026, +0.031]
		F1	0.365 ± 0.011	0.366 ± 0.013	+0.001 [-0.014, +0.016]
Malay	LoRA	F_{faith}	0.311 ± 0.008	0.315 ± 0.008	+0.004 [-0.008, +0.016]
		EM	0.261 ± 0.024	0.234 ± 0.031	-0.027 [-0.057, +0.003]
		F1	0.375 ± 0.014	0.360 ± 0.018	-0.015 [-0.035, +0.006]
Malay	DoRA	F_{faith}	0.314 ± 0.006	0.321 ± 0.013	+0.006 [-0.006, +0.019]
		EM	0.260 ± 0.017	0.237 ± 0.021	-0.023 [-0.045, -0.001]
		F1	0.375 ± 0.008	0.362 ± 0.009	-0.013 [-0.023, -0.004]

pre-specified equivalence margin of ±0.01. The same holds for F1 under both controls and for exact match under the shuffled reward ($[-0.006, +0.006]$), while for exact match under λ_1=0 the 90% interval $[-0.013, +0.005]$ extends past the margin. The real reward exceeds both controls by +0.309 ($[+0.290, +0.328]$). The sampling, scoring and policy-gradient machinery therefore contributes nothing measurable on its own; what moves the score is the dependence of the reward on the sampled answer.

In Malay the real reward is above the controls by +0.014 ($[+0.010, +0.019]$) and +0.015 ($[+0.008, +0.022]$), but this is partly because the controls lie slightly below B (shuffled minus B, −0.007, $[-0.010, -0.004]$) and the real-reward minus B interval includes zero. We read the Malay result as a very small effect whose sign agrees with the Arabic direction, not as a demonstrated gain.

B.4 In Arabic the reward is accompanied by much longer answers

Table B5 describes what the reward changed in the generated answers. Under cross-entropy training and under both controls, Arabic answers average 2.8–2.9 words (the gold answers average 3.2) and 84–87% of them occur verbatim in the passage. After reward training they average 16.5 words, 79.6% are more than three times as long as the gold answer, and only 12.5% occur verbatim in the passage (Figure B1, right). The share of an answer's word 4-grams found in the passage falls from 0.86 to 0.63, so the policy did not learn to copy the passage more: it moved to longer, sentence-like answers that the NLI model judges entailed. Length is not the whole story, because the

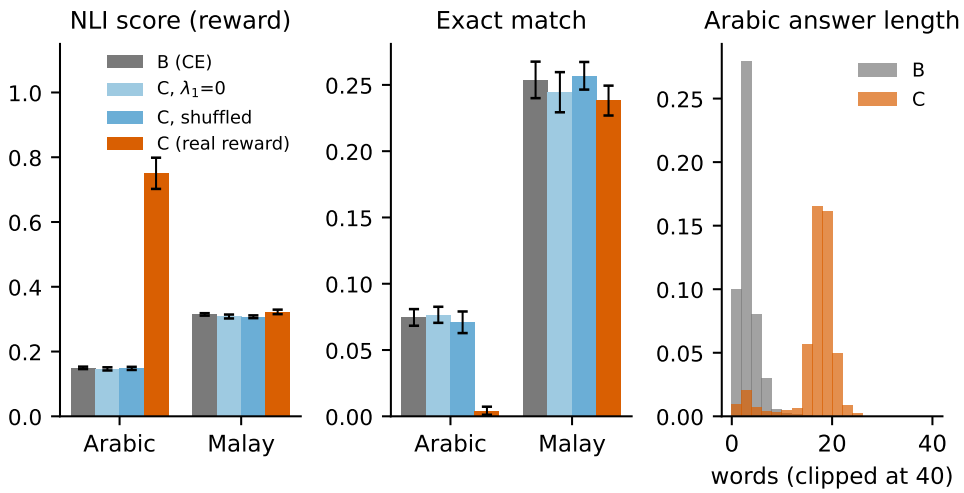


Fig. B1 Corrected study, three adapters pooled. Left and middle: mean NLI score and exact match by language and condition; error bars are the standard deviation over the eight seeds of the adapter-averaged value. Right: distribution of Arabic answer length in words for Config B and Config C (held-out generations, all seeds and adapters; lengths above 40 are clipped).

Table B3 Reward attribution (eight seeds; mean ± SD of F_{faith}). C: real NLI reward; λ_1=0: the sampling and scoring path runs with zero weight on the policy-gradient term; shuffled: rewards permuted across the batch before advantages are formed. Contrast columns give the pooled seed-paired difference (95% CI) over the three adapters within a language.

Lang	**Adapter**	B	λ_1=0	Shuffled	C (real)
Arabic	QLoRA	0.149 ± 0.004	0.146 ± 0.005	0.143 ± 0.007	0.766 ± 0.084
Arabic	LoRA	0.151 ± 0.006	0.148 ± 0.009	0.150 ± 0.005	0.720 ± 0.105
Arabic	DoRA	0.149 ± 0.006	0.146 ± 0.009	0.152 ± 0.009	0.765 ± 0.073
Arabic	pooled: (λ_1=0)−B -0.003 [-0.008, +0.001]; shuffled−B -0.002 [-0.008, +0.004] C−(λ_1=0) +0.604 [+0.565, +0.643]; C−shuffled +0.602 [+0.563, +0.642]				
Malay	QLoRA	0.318 ± 0.006	0.312 ± 0.007	0.308 ± 0.004	0.331 ± 0.012
Malay	LoRA	0.311 ± 0.008	0.311 ± 0.010	0.307 ± 0.008	0.315 ± 0.008
Malay	DoRA	0.314 ± 0.006	0.302 ± 0.015	0.307 ± 0.010	0.321 ± 0.013
Malay	pooled: (λ_1=0)−B -0.006 [-0.014, +0.001]; shuffled−B -0.007 [-0.010, -0.004] C−(λ_1=0) +0.014 [+0.010, +0.019]; C−shuffled +0.015 [+0.008, +0.022]				

8.7% of reward-trained answers of eight words or fewer still score 0.26 against 0.14 for the cross-entropy answers, and length and the score are positively correlated within the reward-trained outputs ($r = 0.27$, against 0.22 for cross-entropy). Exact match falls to 0.004 because an answer that is a full sentence rarely equals a three-word gold span.

Length accounts for most of the Arabic gain, though not all of it (Table B4). Within each of five word-length bins, reward-trained answers score above cross-entropy answers (for example 0.20 against 0.13 for answers of one to three words), but the

Table B4 Arabic NLI score by answer length (words), core stage, three adapters and eight seeds pooled (held-out answers; counts are answers). Mean: mean F_{faith} of the answers in the bin. Matched pairs: answers of B and C to the same item, adapter and seed that fall in the same bin; the difference is C minus B with a 95% passage-cluster bootstrap interval, reported only for bins with at least 30 pairs. Only 7.1% of the 17,184 pairs fall in the same bin, because the two conditions rarely produce answers of similar length.

Words	n_B	Mean B	n_C	Mean C	Matched pairs	C−B [95% CI]
1-3	13,025	0.125	1,028	0.201	959	+0.040 [+0.017, +0.067]
4-6	3,365	0.195	305	0.453	102	+0.376 [+0.162, +0.545]
7-10	658	0.300	241	0.629	29	n/a
11-20	136	0.647	14,719	0.792	131	+0.319 [+0.066, +0.749]
21+	0	–	891	0.839	0	n/a

two conditions seldom produce answers of the same length: 91% of reward-trained answers have 11 words or more against 0.8% of cross-entropy answers, and only 7% of the paired B and C answers to the same item, adapter and seed fall in the same bin. In those pairs the reward-trained answer scores higher by +0.04 ($[+0.02, +0.07]$) for one to three words, +0.38 ($[+0.16, +0.54]$) for four to six and +0.32 for 11 to 20 words (interval $[+0.07, +0.75]$, 131 pairs). A regression of the score on a condition indicator with length-bin fixed effects gives a reward coefficient of +0.14 ($[+0.07, +0.21]$, passage-cluster bootstrap), against +0.60 without the length terms, so adjusting for length bins reduces the estimated gain by about three quarters. This is a descriptive decomposition, not an intervention. The bins are coarse, the long cross-entropy answers are few and selected (they are the ones that happen to copy a passage sentence), and the matched pairs are not representative of all items. We therefore treat length as the main, but not the only, correlate of the gain and do not claim to have isolated a causal route; a length-penalized or length-capped reward would be the direct test and was not run.

In Malay none of these quantities changes: answer length (about 4.0 words), the share of long answers (0.8–1.0%) and the passage overlap are the same in all four conditions (Table B5). The Malay prompt contains the four options, which constrains the output, and the model starts from an exact-match level of about 0.25. We therefore do not read Malay as evidence that the reward is harmless, only that it did not find the same route there. These are descriptive statistics of generations; Appendix B.7 reports what human raters made of them, and Appendix I shows five randomly drawn items with all scores.

B.5 Evaluation state and gradient checkpointing

Each corrected run also re-scores its adapter in the first study's state: the model left in training mode, with dropout active, when held-out answers are generated. Averaged over the core runs, that state lowers the NLI score relative to the headline evaluation by 0.020 for B, 0.016 for λ_1=0, 0.019 for the shuffled reward and 0.302 for the real reward. With gradient checkpointing enabled during training (seeds 42, 123 and 2026), the headline score is unchanged to the reported precision for both B and C (maximum absolute difference 0 in F_{faith}, EM, F1 and option accuracy; Table B6),

Table B5 What the reward changed in the generated answers (eight seeds, three adapters pooled; mean over runs). Words: mean answer length in words (gold answers average 3.2 words in Arabic). Long: share of answers longer than three times the gold answer. In context: share of answers that occur verbatim in the passage. 4-gram: mean share of an answer's word 4-grams found in the passage.

Lang	**Cond.**	F_{faith}	EM	F1	Words	Long	In context	4-gram
Arabic	B	0.150	0.075	0.129	2.9	0.073	0.867	0.857
	λ_1=0	0.146	0.077	0.136	2.8	0.070	0.853	0.846
	Shuffled	0.148	0.071	0.126	2.9	0.071	0.842	0.838
	C (real)	0.750	0.004	0.089	16.5	0.796	0.125	0.632
Malay	B	0.315	0.254	0.372	4.0	0.009	0.259	0.154
	λ_1=0	0.308	0.244	0.368	4.0	0.009	0.255	0.155
	Shuffled	0.307	0.257	0.378	4.0	0.008	0.257	0.155
	C (real)	0.322	0.238	0.363	4.1	0.010	0.248	0.151

Table B6 Evaluation state and gradient checkpointing (seeds 42, 123, 2026; three adapters pooled; mean F_{faith}). Headline: held-out generation in evaluation mode. Legacy state: the same adapter re-scored with the model left in training mode, as in the first study. Checkpointing off/on is the training setting. With checkpointing on, the headline score is unchanged to the reported precision; in the legacy state it collapses for both objectives.

Lang	**Objective**	Ckpt off: headline	Ckpt off: legacy	Ckpt on: headline	Ckpt on: legacy
Arabic	B	0.149	0.134	0.149	0.047
	C	0.749	0.172	0.749	0.038
Malay	B	0.314	0.285	0.314	0.053
	C	0.322	0.300	0.322	0.049

whereas the same adapters in the legacy state fall to 0.04–0.05 in both languages and for both objectives. Checkpointing is therefore neutral for training and for evaluation once evaluation runs in evaluation mode, and the collapse of the first study is an evaluation-state artifact, as Appendix C reproduced on pretrained mT5-base.

The legacy state also hides the reward effect. For the three seeds with checkpointing off, Arabic Config C scores 0.749 in evaluation mode and 0.172 in the legacy state, against 0.149 and 0.134 for Config B; the Arabic contrast is +0.60 in the first and +0.04 in the second. The first study's small checkpointing-off difference (+0.017) is therefore not an estimate of the same quantity as Table B2. We cannot say how much of that gap is the evaluation state and how much is the training-premise and dropout defects of that path (Appendix A.3), because the stage that crosses the dropout policy with the other changes was not run.

B.6 An independent NLI scorer agrees

To check that the Arabic gain is not specific to the reward model, we re-scored every held-out answer of the core and control runs with a different NLI model (`joeddav/xlm-roberta-large-xnli`, a different architecture trained on different data; answers deduplicated by passage and text). The pooled six-cell differences, computed as for the primary contrast, are +0.257 ([+0.240, +0.273]) for C minus B, +0.260

Table B7 Blinded human judgments of the corrected outputs (LoRA, seed 42; 60 source items per language; 248 distinct item–answer judgments rated by each of two annotators). Support: 0 (not supported), 1 (partly), 2 (fully supported by the passage). Correct: share of answers judged correct. Δ: C minus B on support, with a 95% item-level bootstrap interval. The mean row averages the two annotators per answer.

Lang	**Rater**	Support B	Support C	Δ support [95% CI]	Correct B	Correct C
Arabic	Annotator 1	1.67	1.17	−0.50 [−0.70, −0.30]	0.88	0.90
	Annotator 2	1.85	1.92	+0.07 [−0.05, +0.20]	0.95	1.00
	Mean	1.76	1.54	−0.22 [−0.36, −0.08]	0.92	0.95
Malay	Annotator 1	0.82	0.87	+0.05 [−0.12, +0.22]	0.22	0.17
	Annotator 2	1.73	1.75	+0.02 [−0.08, +0.10]	0.92	0.95
	Mean	1.27	1.31	+0.03 [−0.10, +0.15]	0.57	0.56

([+0.243, +0.277]) against λ_1=0 and +0.258 ([+0.242, +0.275]) against the shuffled reward (eight seeds), against +0.304, +0.309 and +0.309 under the reward model. The two scorers correlate at Spearman $\rho = 0.55$ over all answers. The effect is therefore not an artifact of a single NLI model, although the size is smaller and the pooled value is dominated by the Arabic cells. Because both scorers are entailment classifiers, agreement shows that sentence-length answers are judged entailed by two models; it does not show that they answer the question. Exact match and F1 are low for these answers, but they penalize any answer that is longer than the reference span, so they do not settle the question either; the human ratings below address it directly.

B.7 Blinded human judgments do not follow the NLI score

Two annotators, each blind to the training condition, rated the answers of the LoRA adapter at seed 42 for 60 randomly drawn source items per language (identical answers from different conditions shown once, 248 distinct judgments per annotator). They scored whether the passage supports the answer (0–2) and whether the answer is correct (0/1); the annotation protocol is in the repository and the annotators were volunteers (Statements and Declarations). Agreement is limited: quadratic-weighted Cohen's $\kappa = 0.38$ for support and 0.18 for correctness. Exact agreement on support is 57% in Arabic and 20% in Malay. The raters also differ in calibration (Table B7): Annotator 2 judges almost every answer fully supported and correct, and Annotator 1 is stricter. Because exact match undercounts valid longer answers, a disagreement with exact match alone does not show that a rater is wrong, so we checked Malay correctness against an independent label: Belebele provides a gold option for every question, and 23% of the 81 distinct Malay answers select it. Annotator 1 marks 20% of these answers correct (agreement with the option label 67%, Cohen's $\kappa = 0.02$) and Annotator 2 marks 91% correct (agreement 25%, $\kappa = -0.05$); neither rater's Malay correctness labels are informative about the gold option beyond chance. We keep both annotators and their mean in all tables and draw no Malay conclusion from human correctness. Annotator 2's Arabic ratings are near the ceiling (support is identical for the B and C answer of 53 of 60 items), which limits their power to detect a difference; they do not show that no difference exists.

In Arabic, the mean human support of reward-trained answers is lower than that of cross-entropy answers (1.54 against 1.76; $\Delta = -0.22$, 95% CI [−0.36, −0.08]), while

the zero-weight and shuffled-reward answers are not (1.79 and 1.77). The difference is carried by Annotator 1 (-0.50, $[-0.70, -0.30]$; the C answer is rated lower for 32 of 60 items, equal for 22 and higher for 6); Annotator 2 sees none ($+0.07$, $[-0.05, +0.20]$; lower for 3, equal for 53, higher for 4). Taking the lower of the two ratings for each answer gives -0.50 ($[-0.68, -0.32]$). The Spearman correlation between the reward-model score and mean human support is -0.30 in Arabic and $+0.12$ in Malay (all items pooled within language). Human-judged correctness does not change in Arabic ($0.92 \rightarrow 0.95$ on the mean; $0.88 \rightarrow 0.90$ for Annotator 1), so the near-zero exact match of the long answers reflects their form, a full sentence in place of a short span, and not necessarily wrong content. In Malay no human outcome differs between B and C (Δ support $+0.03$, $[-0.10, +0.15]$). The human evidence therefore gives no support for a gain in faithfulness from the reward. It is tentative evidence of a loss in Arabic: one of two annotators sees it, the other's ratings are too close to the ceiling to detect it, and neither sees a change in correctness. It does not show that the answers are wrong. With one adapter, one seed, 60 items per language and two raters who disagree, these results are a validity check on the metric and not a precise estimate of the reward's effect on human-judged quality; a stronger test needs more qualified raters, adjudication of disagreements and more adapters and seeds (Section 6).

B.8 Summary of evidence

The corrected study supports five statements. First, the first study's headline harm is not a reward effect: it follows the evaluation state and gradient checkpointing, and disappears when evaluation runs in evaluation mode. Second, the NLI reward changes the generated answers: with eight seeds and two null controls (zero weight and shuffled rewards), the Arabic NLI score rises by about 0.6, and the controls rule out the training machinery as the cause. Third, this rise diverges from the other measures: exact match and F1 fall in Arabic, the answers become about six times longer, and adjusting for answer-length bins reduces the estimated gain by about three quarters; human raters judge the answers no more correct and, on average, less supported (one of two raters sees the loss), which we read as consistent with over-optimization of the NLI score but do not treat as demonstrated. In Malay the NLI score rises by a small, imprecise amount ($+0.008$), answer length and passage overlap do not change, and exact match falls slightly (exploratory, -0.016); no human outcome differs between B and C. Fourth, an independent NLI scorer reproduces the gain at a smaller size ($+0.257$ against $+0.304$). Fifth, the evidence has limits: human validation is small (60 items per language, two raters with limited agreement, one adapter and seed) and the study covers one backbone and one reward weight.

Appendix C Results of the First (Pre-Correction) Study

This appendix keeps the results of the first study: 188 runs on the pre-correction training path (three seeds, one GPU model and software stack, none failed), which preceded the audit of Appendix A.3. They are retained because they document how the evaluation-state artifact arose and what it looked like; they are exploratory, were

Table C8 mT5-base, pre-correction training path (Sections A.2 and A.3), seeds 42, 123 and 2026; mean (SD) over seeds. Held-out generation ran in the evaluation state described in Section C.1. B: CE-only LoRA. C(off): composite objective, gradient checkpointing off. C(default): composite objective under the checkpointing setting of the original protocol (on for every non-quantized variant, off for QLoRA, so QLoRA's C(default) is the same run as C(off)). Held-out examples: 716 Arabic (720 minus 4 whose passage also occurs in the training slice) and 430 Malay. For AdaLoRA and VeRA, C(off) has seeds 123 and 2026 only ($n = 2$).

		F_{faith}			EM		
Lang	**Variant**	B	C(off)	C(default)	B	C(off)	C(default)
Arabic	QLoRA	0.140 (0.015)	0.141 (0.003)	= C(off)	0.013 (0.005)	0.015 (0.004)	= C(off)
Arabic	LoRA	0.133 (0.002)	0.161 (0.004)	0.042 (0.007)	0.013 (0.003)	0.015 (0.004)	0.002 (0.001)
Arabic	DoRA	0.130 (0.003)	0.147 (0.007)	0.051 (0.004)	0.022 (0.006)	0.016 (0.005)	0.002 (0.001)
Arabic	AdaLoRA	0.065 (0.001)	0.067 (0.000)	0.064 (0.002)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
Arabic	VeRA	0.070 (0.000)	0.067 (0.000)	0.062 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
Malay	QLoRA	0.286 (0.007)	0.309 (0.018)	= C(off)	0.151 (0.013)	0.152 (0.020)	= C(off)
Malay	LoRA	0.275 (0.011)	0.295 (0.008)	0.048 (0.007)	0.171 (0.022)	0.130 (0.030)	0.007 (0.004)
Malay	DoRA	0.295 (0.019)	0.310 (0.011)	0.054 (0.004)	0.149 (0.026)	0.133 (0.027)	0.009 (0.001)
Malay	AdaLoRA	0.039 (0.007)	0.037 (0.000)	0.027 (0.001)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
Malay	VeRA	0.032 (0.000)	0.037 (0.000)	0.032 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)

produced by a path with the defects of Table A1, and are superseded by the corrected study of Appendix B. In particular, the small checkpointing-off difference reported in Appendix C.2 is not comparable to the corrected estimate (Appendix B.5). Table C8 gives cell means for every adapter, including AdaLoRA and VeRA, which were not re-run.

C.1 The original comparison reproduces, and it follows the evaluation state

Under the original configuration policy the composite objective lowers F_{faith} for the two non-quantized variants that were checkpointed: $\Delta = -0.160$ for DoRA ($p = 0.001$) and -0.159 for LoRA ($p = 0.001$), negative in all 12 variant-language-seed cells of the two variants, and $+0.012$ for QLoRA, which was not checkpointed (95% CI $[-0.033, +0.057]$; Table C12, last block). Exact match falls from 0.15–0.17 to under 0.01 in Malay for these variants (Table C8, C(default)). This is the pattern our initial analysis attributed to the variant: QLoRA improves, DoRA and LoRA collapse.

Table C9 shows that the pattern follows the checkpointing flag, not the reward or the adapter. At seed 42, with checkpointing on, Config C collapses to $F_{\text{faith}} \approx 0.03$–$0.06$ for LoRA and DoRA in both languages when the reward weight is zero (λ_1=0) and when the reward is shuffled. For QLoRA, whose control runs were never checkpointed, nothing collapses; forcing checkpointing on for QLoRA Config C produces the same collapse (Arabic 0.045, Malay 0.054, against 0.141 and 0.309 with it off). With checkpointing off, the identical code for all three adapters returns to the level of Config B or above (Table C8, C(off)). Across the six cells, turning checkpointing off raises Config C by $+0.178$ on average (six of six positive). One seed per cell suffices because the effect is about ten times the seed-to-seed spread.

Mechanism. Checkpointing is meant to change memory and speed only. We found that in the T5 code path of the library versions we used, it also changes what `generate()` computes. A model in training mode with gradient checkpointing enabled

Table C9 Gradient-checkpointing audit, seed 42 (one run per cell). F_{faith} of Config C with checkpointing on and off, and of the two controls with checkpointing on, against Config B. The two control columns follow the original protocol's default checkpointing policy: on for LoRA and DoRA, off for QLoRA (so the QLoRA entries of these two columns were run with checkpointing emphoff). With checkpointing on, Config C collapses whether the reward is real, shuffled or absent (λ_1=0), and QLoRA collapses too when checkpointing is forced on for it; with checkpointing off the same code does not collapse.

Lang	Variant	B	C, ckpt on	C, ckpt off	λ_1=0, on	Shuffled, on
Arabic	QLoRA	0.1366	0.0446	0.1386	0.1300	0.1367
Arabic	LoRA	0.1350	0.0343	0.1566	0.0503	0.0399
Arabic	DoRA	0.1319	0.0470	0.1451	0.0498	0.0517
Malay	QLoRA	0.2797	0.0538	0.3237	0.2865	0.2975
Malay	LoRA	0.2869	0.0467	0.2889	0.0511	0.0416
Malay	DoRA	0.2739	0.0585	0.2970	0.0481	0.0541

Table C10 Decoding check on pretrained `google/mt5-base` with a randomly perturbed LoRA adapter (rank 8), Transformers 5.5.0, PEFT 0.21.2, PyTorch 2.14.1 (CPU), all dropout probabilities set to zero, greedy decoding, two prompts. "Identical" means the generated token sequences agree up to the first end-of-sequence token. Last-token-only and full-prefix decoders are explicit loops written for this check.

Mode of the model during `generate()`	Compared with	Identical
training, checkpointing off	evaluation mode	yes
training, checkpointing on	evaluation mode	no
evaluation mode	full-prefix decoder	yes
training, checkpointing on	last-token-only decoder	yes

discards the key–value cache, but the generation loop still feeds the decoder only the most recent token, so each new token is predicted from the last token alone instead of the full prefix. The held-out evaluation of the pre-correction runs called `generate()` after training without ever switching the model to evaluation mode (Appendix A.3), and Config C was the only configuration in which checkpointing had been enabled. Table C10 reproduces the effect on pretrained `google/mt5-base` with the paper's library versions: with dropout disabled everywhere, training-mode decoding without checkpointing matches evaluation-mode decoding, while training-mode decoding with checkpointing matches a last-token-only decoder and differs from the full-prefix decoder. A minimal test on a tiny T5 shows the same behavior for sampling, together with a mismatch of 17–22 nats between the log-probability that `generate()` assigns to a sample and the one the scorer recomputes whenever dropout is active in training mode (and agreement to 10^{-5} in evaluation mode) (`ckpt_repro.py`, `tests_v2.py`).

This account fits all four observations: Config B (never checkpointed) is unaffected; Config C collapses only where checkpointing was on; the collapse persists at λ_1=0 and under shuffled rewards because those runs share the evaluation state; and QLoRA collapses when checkpointing is forced on. It does not yet establish how much of the collapse arose in training, where the policy-gradient sampling of a checkpointed run passed through the same defective path, and how much in evaluation. At λ_1=0 the

Table C11 Reward-attribution controls (gradient checkpointing off; three seeds; mean ± SD of F_{faith}). λ_1=0: the sampling-and-scoring path runs but the policy-gradient term has zero weight. Shuffled: the NLI rewards are permuted across the batch before advantages are computed, which preserves the reward distribution but removes its dependence on the sampled answer. B is repeated from Table C8.

Lang	Variant	B	λ_1=0	Shuffled	Real reward
Arabic	QLoRA	0.1403 ± 0.0152	0.1349 ± 0.0044	0.1339 ± 0.0033	0.1414 ± 0.0027
Arabic	LoRA	0.1334 ± 0.0016	0.1332 ± 0.0021	0.1257 ± 0.0007	0.1614 ± 0.0043
Arabic	DoRA	0.1298 ± 0.0027	0.1369 ± 0.0035	0.1286 ± 0.0168	0.1470 ± 0.0069
Malay	QLoRA	0.2859 ± 0.0072	0.2893 ± 0.0146	0.2877 ± 0.0104	0.3093 ± 0.0177
Malay	LoRA	0.2748 ± 0.0107	0.2857 ± 0.0155	0.2873 ± 0.0279	0.2949 ± 0.0082
Malay	DoRA	0.2955 ± 0.0188	0.2733 ± 0.0139	0.2826 ± 0.0105	0.3098 ± 0.0112

policy-gradient term has no weight, so training there reduces to checkpointed cross-entropy, which points to evaluation, but we saved no adapter weights and cannot re-evaluate the same adapter in the other state. The corrected study separates the two by crossing checkpointing with the cross-entropy objective and by re-scoring every trained adapter in both evaluation states (Appendix B.5): checkpointing is neutral in training, and the collapse is an evaluation-state effect. The legacy T5Gemma 2 runs (Appendix F) used the same policy and show no collapse; whether a backbone is affected depends on its `generate()` implementation, which we did not test there. The scientific consequence is narrow and firm: the original contrast between Config C and Config B was a contrast between two evaluation-and-training states, and its sign could be reversed by a memory flag.

C.2 With checkpointing off, a small and imprecise difference remains

With checkpointing off for all adapters, Config C exceeds Config B in all six adapter-language means (Table C8) and in 16 of the 18 individual seed cells. The language-averaged differences are +0.016 (DoRA), +0.024 (LoRA) and +0.012 (QLoRA); their 95% confidence intervals with three seeds are wide, and after Holm correction over the three adapters none is below 0.05 (Table C12). Pooled over the three adapters the difference is +0.017 with a 95% confidence interval of $[-0.003, +0.038]$, which includes zero (paired $p = 0.069$). For DoRA and LoRA it is +0.017 to +0.028 in Arabic and +0.014 to +0.020 in Malay; for QLoRA it is +0.001 in Arabic and +0.023 in Malay.

Table C11 and the lower blocks of Table C12 report the controls. Neither control departs from Config B: λ_1=0 differs from B by −0.001 (95% CI $[-0.015, +0.013]$) and the shuffled reward by −0.002 ($[-0.025, +0.020]$). The real reward exceeds the λ_1=0 control by +0.018 (95% CI $[+0.008, +0.029]$; 17 of 18 cells positive) and the shuffled-reward control by +0.020 ($[-0.007, +0.046]$; 16 of 18), so the second interval includes zero. In each adapter the real reward is above both controls on average, but no single-adapter interval excludes zero except DoRA against cross-entropy, and that one does not survive correction. The direction is consistent. The evidence does not allow us to say that the reward improves the entailment score, and it is further limited because

Table C12 Language-averaged contrasts in F_{faith}, seed-paired ($n = 3$ seeds, 2 degrees of freedom). p: two-sided paired t; p_{Holm}: Holm correction over the three variants within a contrast. Cells: variant-by-language-by-seed differences that are positive (descriptive; the 18 cells are not independent). "All" pools the three variants (per-seed mean over six cells; 95% CI shown); it treats variants as exchangeable and is descriptive.

Contrast	Variant	Δ	p	p_{Holm}	Cells > 0	95% CI
C(off) − B	QLoRA	+0.0122	0.362	0.362	4/6	[−0.033, +0.057]
	LoRA	+0.0240	0.073	0.146	6/6	[−0.005, +0.053]
	DoRA	+0.0158	0.025	0.076	6/6	[+0.005, +0.027]
	All	+0.0173	0.069	–	16/18	[−0.003, +0.038]
C(off) − (λ_1=0)	QLoRA	+0.0132	0.114	0.233	6/6	[−0.008, +0.034]
	LoRA	+0.0186	0.104	0.233	5/6	[−0.010, +0.047]
	DoRA	+0.0233	0.078	0.233	6/6	[−0.006, +0.053]
	All	+0.0184	0.016	–	17/18	[+0.008, +0.029]
C(off) − shuffled	QLoRA	+0.0146	0.122	0.367	6/6	[−0.010, +0.039]
	LoRA	+0.0216	0.201	0.367	4/6	[−0.028, +0.071]
	DoRA	+0.0228	0.127	0.367	6/6	[−0.016, +0.062]
	All	+0.0197	0.086	–	16/18	[−0.007, +0.046]
(λ_1=0) − B	QLoRA	−0.0010	0.928	0.928	3/6	[−0.043, +0.041]
	LoRA	+0.0054	0.237	0.712	4/6	[−0.009, +0.019]
	DoRA	−0.0076	0.296	0.712	3/6	[−0.031, +0.016]
	All	−0.0011	0.765	–	10/18	[−0.015, +0.013]
shuffled − B	QLoRA	−0.0023	0.756	1.000	3/6	[−0.030, +0.025]
	LoRA	+0.0024	0.738	1.000	2/6	[−0.024, +0.029]
	DoRA	−0.0071	0.593	1.000	3/6	[−0.056, +0.041]
	All	−0.0023	0.703	–	8/18	[−0.025, +0.020]
C(default) − B	QLoRA	+0.0122	0.362	0.362	4/6	[−0.033, +0.057]
	LoRA	−0.1590	0.001	0.003	0/6	[−0.185, −0.133]
	DoRA	−0.1602	0.001	0.003	0/6	[−0.182, −0.139]
	All	−0.1023	0.001	–	4/18	[−0.114, −0.090]

this path scored a different premise in training than in evaluation and sampled with dropout on (Appendix A.3): it measures a defective version of the intended objective.

C.3 Correctness and the lower-scoring adapters

The difference in the entailment score does not extend to correctness. With checkpointing off, Arabic exact match is near zero everywhere (at most 0.022) and unchanged by the objective (Table C8). In Malay it is unchanged for QLoRA (0.151 → 0.152) and lower for DoRA (0.149 → 0.133) and LoRA (0.171 → 0.130). Token F1 (Table C13) shows the same absence of improvement: Malay F1 is 0.309 → 0.322 for QLoRA, 0.310 → 0.302 for DoRA and 0.334 → 0.304 for LoRA, and Arabic F1 stays below 0.06 in every condition. The shuffled reward gives a similar Malay exact-match loss for LoRA (0.140) while λ_1=0 does not (0.169), so we cannot tell whether this cost belongs to the reward signal or to the sampling path. Mean mT5 Arabic exact match stays an order of magnitude below Malay's in every condition, so mT5-base does not solve the Arabic task in this setting; the corrected study did not add a frozen or fully fine-tuned reference.

AdaLoRA and VeRA remain at the floor under both objectives (EM = 0.000; $F_{\text{faith}} \in [0.03, 0.07]$). Their differences between B and C(off) are at most 0.006 and

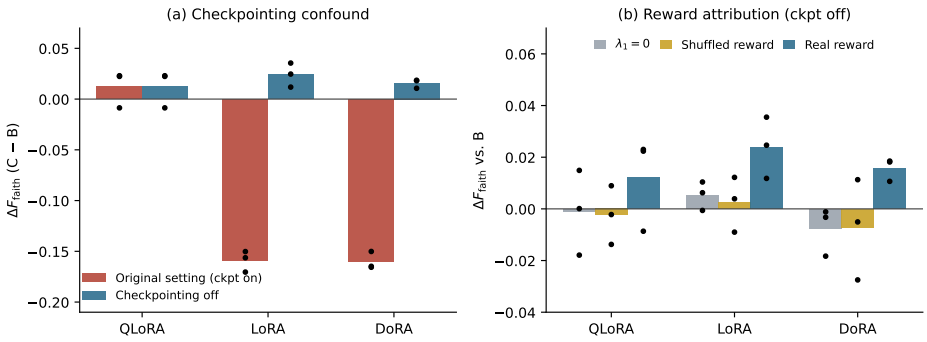


Fig. C2 Language-averaged ΔF_{faith} by adapter; bars are the mean over three seeds and dots the individual seeds. (a) Config C minus Config B under the original checkpointing setting (reproducing the original "harm") and with checkpointing off; QLoRA's two bars coincide because it was never checkpointed. (b) With checkpointing off, the real reward, the shuffled reward and λ_1=0, each minus Config B.

Table C13 Token F1 (mean ± SD over seeds), same runs and conventions as Table C8. Token F1 is independent of the NLI reward.

Lang	**Variant**	B	C(off)	C(default)
Arabic	QLoRA	0.047 ± 0.005	0.049 ± 0.003	same as C(off)
Arabic	LoRA	0.048 ± 0.007	0.052 ± 0.008	0.003 ± 0.003
Arabic	DoRA	0.055 ± 0.012	0.052 ± 0.015	0.005 ± 0.002
Arabic	AdaLoRA	0.001 ± 0.000	0.000 ± 0.001	0.000 ± 0.000
Arabic	VeRA	0.001 ± 0.000	0.001 ± 0.000	0.000 ± 0.000
Malay	QLoRA	0.309 ± 0.004	0.322 ± 0.016	same as C(off)
Malay	LoRA	0.334 ± 0.010	0.304 ± 0.013	0.077 ± 0.015
Malay	DoRA	0.310 ± 0.013	0.302 ± 0.020	0.083 ± 0.010
Malay	AdaLoRA	0.012 ± 0.004	0.008 ± 0.001	0.004 ± 0.002
Malay	VeRA	0.003 ± 0.000	0.003 ± 0.000	0.000 ± 0.000

have no interpretation; VeRA's outputs are identical across seeds (zero SD). In the pre-correction runs AdaLoRA also never reallocated its rank (Appendix A.3), so it behaved as a fixed rank-12 adapter. We therefore do not test the reward on them, and their C(default) rows (down to 0.027 in Malay for AdaLoRA) should not be read as a checkpointing effect, since their baseline is already at the floor.

C.4 Repeated runs and the same-seed discrepancy

Our initial analysis contained two runs of the same condition and seed that differed in F_{faith}, and the hardware for those runs was recorded inconsistently. We repeated DoRA/Malay and QLoRA/Malay (seed 42, Configs B and C) four times each on the same GPU model and software stack: twice with default kernels and twice with deterministic algorithms and a fixed cuBLAS workspace (Table C14). All four runs of every cell agree to the reported four decimals in F_{faith} and exactly in exact match, whether or not deterministic kernels are enabled. We did not compare model weights or per-example outputs bit by bit. On a fixed stack we therefore found no run-to-run variation

Table C14 Repeated runs, seed 42, Malay, F_{faith} (EM in parentheses). nd: default kernels; det: deterministic algorithms. Config C was run under the original checkpointing policy.

Variant	Config	nd1	nd2	det1	det2
DoRA	B	0.2739 (0.174)	0.2739 (0.174)	0.2739 (0.174)	0.2739 (0.174)
DoRA	C	0.0585 (0.009)	0.0585 (0.009)	0.0585 (0.009)	0.0585 (0.009)
QLoRA	B	0.2797 (0.137)	0.2797 (0.137)	0.2797 (0.137)	0.2797 (0.137)
QLoRA	C	0.3237 (0.174)	0.3237 (0.174)	0.3237 (0.174)	0.3237 (0.174)

at the reported precision, and the earlier discrepancy was not reproduced. We cannot say what caused it; the two earlier runs were recorded on different or unrecorded environments, and every run record now stores its GPU, driver and library versions. The repeats support reporting the seed-to-seed spread in Table C8 as variation due to the seed, but they do not imply identical results across hardware or library versions.

C.5 What the corrected study showed

The pre-correction runs supported three statements: the original harm is a configuration effect that appears without the reward and disappears when checkpointing is off; with that confound removed, Config C was slightly above Config B in the NLI score ($\approx +0.02$, pooled interval including zero); and the difference did not improve exact match or F1. The corrected study (Appendix B) confirmed the first and third statements and showed that the second was a misleading estimate: in evaluation mode with a correct premise and a dropout-free policy, the reward raises the Arabic NLI score by about 0.6 and the Malay score by about 0.01 (Table B2), and the first study's evaluation state hid most of that difference (Appendix B.5).

Appendix D Implementation Validation and Corrected Training Objective

This appendix documents the training objective and two corrections to it. The composite objective's faithfulness term (Equation (1)) requires a valid gradient estimator or differentiable surrogate for F_{faith}, since discrete decoding is not itself differentiable. The initial version of the training code violated this requirement: the original implementation computed the batch's generation and its faithfulness score entirely under `torch.no_grad()`, then added the resulting *Python float* to the scalar cross-entropy loss:

```
with torch.no_grad():
    gen_ids = model.generate(input_ids=...,
                             attention_mask=...)
pred_text = tokenizer.decode(gen_ids)
score = f_faith(pred_text, context_text)  # float
loss = ce_loss + lambda1 * (1.0 - score)
```

Because $\mathrm{d}(\mathcal{L}_{\text{CE}} + c)/\mathrm{d}\theta = \mathrm{d}\mathcal{L}_{\text{CE}}/\mathrm{d}\theta$ for any constant c and parameter θ, this implementation's backward pass was *mathematically identical* to Config B's for every trainable parameter: the faithfulness term changed the number written to the training log but altered no optimizer step. We confirmed this with a controlled test comparing gradients from (a) cross-entropy loss alone and (b) that loss plus a constant

offset standing in for the faithfulness term, backpropagated from the identical forward pass (`eval()` mode, removing dropout as a confound): the two gradient sets were bit-identical (maximum absolute difference 0.00×10^0). The original script also set no random seed, so this implementation's two configurations were, in effect, two independently-initialized replicates of the identical cross-entropy-only procedure.

The fix. We replace the non-differentiable path with a policy-gradient objective in the self-critical family of Rennie et al (2017), but with a leave-one-out sample baseline instead of that paper's greedy-decoding baseline; formally it is REINFORCE with a leave-one-out baseline (Ahmadian et al, 2024), and we use "SCST-style" only as a short label. For each example, we sample K=4 generations from the current policy ($\hat{y}_1, \ldots, \hat{y}_K$, `do_sample=True`, temperature 1.0, no truncation; sampling is non-differentiable by construction and performed under `no_grad`, as standard for policy-gradient methods), score each with the frozen NLI reward model F_{faith} (also `no_grad`), and compute a leave-one-out advantage for sample k,

$$A_k = F_{\text{faith}}(\hat{y}_k) - \frac{1}{K-1} \sum_{j \neq k} F_{\text{faith}}(\hat{y}_j), \tag{D1}$$

a detached scalar weight. We then recompute $\log p_\theta(\hat{y}_k \mid x)$ via a *separate*, differentiable, teacher-forced forward pass using $\hat{y}_k$ as the decoder target, the only quantity in the objective depending on θ (the LoRA adapter parameters; the frozen backbone receives no gradient by construction) and where gradient information enters the graph. The corrected objective is

$$\mathcal{L}_G = \mathcal{L}_{\text{CE}} - \frac{\lambda_1}{K} \sum_{k=1}^{K} A_k \log p_\theta(\hat{y}_k \mid x), \quad \lambda_1 = 0.3, \tag{D2}$$

the standard REINFORCE / score-function gradient estimator $\nabla_\theta \mathbb{E}[R] \approx (R - b)\nabla_\theta \log p_\theta(\hat{y})$ with a leave-one-out baseline b for variance reduction. Two properties of the estimator matter for correctness, and our initial implementation violated both. First, the sampling distribution must equal the distribution being scored. The original sampler used nucleus truncation (top-p 0.95) while the scorer used the full softmax, so the samples came from $q \neq p_\theta$ and the estimator was off-policy without importance weights. The corrected path samples with `top_p=1.0`, `top_k=0` and sentinel tokens suppressed, and the scorer applies the same suppression (banned logits set to -10^9) after sanitizing NaN or Inf logits, so both use the same renormalized distribution. Second, for the T5 family the teacher-forced pass must condition the decoder exactly as generation did. The original scorer passed the sampled sequence, which already begins with the decoder-start token, as `labels`, so the model shifted it again and the start token was fed twice. The corrected scorer feeds `sampled[:, :-1]` as `decoder_input_ids` and scores `sampled[:, 1:]`, masks only padding (the end-of-sequence token is counted), and asserts that every sampled sequence starts with the decoder-start token. Full pseudocode is in the released training script.

Verification. A CPU unit test (`analysis/estimator_checks_fixed.py`; tiny randomly initialized T5, repository functions extracted verbatim) checks that (F1)

the aligned log-probability equals the log-probability that `generate()` assigned to the sampled sequence (maximum absolute difference 7.6×10^{-6} nats), (F2) the original scorer deviates from generation (mean absolute difference 2.2 nats in this test model; we did not measure the deviation on trained mT5 models), (F3) banned tokens are never sampled and carry zero probability under the scorer, and (F4) gradients are nonzero and a constant reward gives exactly zero advantage and gradient. All checks pass. Earlier tests also confirmed that advantages sum to zero within each group of K samples and that a small descent step on $-A \log p_\theta$ raises $\log p_\theta$ for $A > 0$ and lowers it for $A < 0$. Half of the LoRA parameters (the A matrices) receive exactly zero gradient at the first step because B is zero-initialized; this is expected and B receives nonzero gradient.

Numerical guards. Under QLoRA's 4-bit quantization, sampling crashed on NaN or Inf logits and, in one earlier run, evaluation silently produced empty strings for every example (driving F_{faith} to exactly zero). A shared logit-sanitization guard fixed both. We document the second failure because an unnoticed empty-generation run would have been reported as a spurious "QLoRA has zero faithfulness on Malay" result.

Tests under production conditions (corrected code). The unit test above ran in evaluation mode on a tiny model. The pre-correction training ran sampling and re-scoring in training mode, with dropout drawn independently in the two passes. `tests_v2.py` repeats the check on a PEFT-wrapped model with model and adapter dropout 0.3, in float32 and bfloat16, with gradient checkpointing on and off, in both modes. In evaluation mode the log-probability that `generate()` assigns to a sample and the one the scorer recomputes agree to 1.1×10^{-5} nats in every configuration; in training mode they differ by 17 to 22 nats. The corrected policy branch therefore samples and re-scores with dropout off and restores training mode for the cross-entropy pass. The same file tests that the NLI premise passed to the reward is the raw passage (a synthetic wrong-option marker never reaches it; the legacy path does pass it), that held-out generation runs in evaluation mode (the legacy path does not), that the sanitizer is shared and counted, and that AdaLoRA's number of non-zero singular values falls from 432 to 288 on a toy run when the scheduler is called and stays at 432 when it is not.

What this appendix does not claim. The estimator fixes remove defects in the scoring path. They do not by themselves explain the checkpointing confound of Appendix C.1, which appears at $\lambda_1 = 0$, where the policy-gradient term has no weight; that confound is explained by the decoding state during evaluation.

Appendix E Pipeline Validation Before the Final Experiments

End-to-end validation surfaced three implementation issues, distinct from the non-differentiability bug above, that affected every condition and were fixed before any reported result was produced. (i) *Target-module coverage:* LoRA's `target_modules` initially covered only the attention projections; it now also includes mT5's gated feed-forward sublayers (`wi_0`, `wi_1`, `wo`) (Section 3). (ii) *Sentinel tokens:* mT5's

`<extra_id_N>` tokens were not suppressed, so weakly fine-tuned models could emit span-corruption output; their IDs are now passed to `suppress_tokens` at both training-time sampling and evaluation-time generation (leakage: 0 of 100 logged generations after the fix, 56% before). (iii) *Evaluation slice:* the held-out slice was fixed at 30 examples; every example outside the 470-example training split is now used (720 Arabic, 430 Malay; Section 3.1), a 24-fold expansion for Arabic. Absolute values in this paper should not be compared with any pre-validation figures, including the frozen baseline.

Appendix F First-Protocol Runs on Qwen3 and T5Gemma 2 (Superseded)

Before the primary study, Qwen3-0.6B-Base (four LoRA variants) and T5Gemma 2 270M-270M (three variants) were run under the first protocol with three seeds, the original estimator (including a double-fed decoder-start token on the encoder-decoder backbone and nucleus-truncated sampling scored under the full softmax), the original checkpointing policy and evaluation without `eval()`. Under that protocol the composite objective changed the language-averaged NLI score by between -0.008 and $+0.008$ for the Qwen3 variants (all intervals included zero) and by $+0.0036$, $+0.0005$ and -0.0025 for T5Gemma 2 (QLoRA, DoRA, LoRA; $p \geq 0.60$). These runs are superseded by the primary study of Section 4, which uses the corrected path, five seeds and the two reward-information controls on both backbones, and their tables are not reproduced here; the run records are in the release. The two sets of results agree qualitatively: no large effect of the reward on either backbone.

Appendix G Blinded Human Validation of Original-Protocol Generations

Status. The annotated generations are from the original protocol: seed 42, QLoRA and DoRA, Configs B and C, with the original checkpointing policy, the original estimator, and the original configuration under which DoRA's Config C had collapsed. They were produced before the controls of this paper existed. The validation therefore speaks to the NLI score and to what the collapsed outputs look like, and says nothing about the checkpointing-off gain.

G.1 Protocol

To check the NLI score against human judgment we drew a stratified, de-duplicated sample of 350 held-out mT5 generations (175 per language) from the seed-42 runs of QLoRA and DoRA under Config B and Config C, the two variants on which the central finding rests. Cell sizes are therefore unbalanced (Table G1). Items were shuffled and re-issued anonymous identifiers, and annotators saw only the passage, the question, and the generated answer; the configuration, the LoRA variant, the seed, the NLI score, EM, and F1 were withheld. The reference answer was shown only after the faithfulness judgment (below) had been recorded.

Table G1 Human-judged share of answers fully supported by the passage (final adjudicated Q1 = 2), mT5-base, seed 42, with Wilson 95% intervals, alongside the mean NLI score F_{faith} of the same items. Cell sizes differ because the sample was stratified and de-duplicated.

Lang	Variant	Config	n	Supported	95% CI	NLI
Arabic	DoRA	B (CE)	45	0.578	[0.433, 0.710]	0.148
Arabic	DoRA	C (comp.)	47	0.085	[0.034, 0.199]	0.062
Arabic	QLoRA	B (CE)	34	0.500	[0.341, 0.659]	0.159
Arabic	QLoRA	C (comp.)	49	0.551	[0.413, 0.681]	0.119
Malay	DoRA	B (CE)	82	0.427	[0.325, 0.535]	0.280
Malay	DoRA	C (comp.)	33	0.273	[0.151, 0.442]	0.034
Malay	QLoRA	B (CE)	34	0.441	[0.289, 0.605]	0.298
Malay	QLoRA	C (comp.)	26	0.462	[0.288, 0.645]	0.264

Two independent annotators per language, fluent readers of Modern Standard Arabic and Malay respectively, labelled all 175 items of their language after a 15-item pilot that was identical for both annotators. They were adult volunteers who knew the labels would be used in an academic publication, no personal data were collected beyond the labels (ethics details are in the submission system), and they were instructed to work alone and not to use AI tools or web search. For each item they gave a faithfulness label judged from the passage alone (Q1: 2, fully supported; 1, partly supported; 0, unsupported or contradicted; X, empty or unreadable), a correctness label against the reference (Q2: 2/1/0), and a copy label (Q3: yes/no). Disagreements on Q1 were resolved by a third reading (95 items; 51 Arabic, 44 Malay). The Arabic disagreements were adjudicated by the first author, a native Arabic speaker, who saw both annotator labels but not the condition; the Malay disagreements by a separate, independent Malay reader with the same information. An item is *supported* if its final Q1 label is 2. For ordinal summaries X is scored 0.

We report raw agreement, nominal Cohen's κ over the four Q1 labels, quadratic-weighted κ over 0/1/2 (X as 0), and κ for supported versus not supported. Supported rates carry Wilson 95% intervals; the Config C minus Config B difference in the supported rate uses a normal-approximation interval for two independent proportions. Because the sample is a single seed and the cells are small, these contrasts are checks on the direction and rough size of the NLI-score effects, not replacements for the 3-seed analysis of Appendix A.5. As a sensitivity check we repeat each contrast using each annotator's unadjudicated labels. The copy label (Q3) is not analysed, because its inter-annotator agreement was near zero ($\kappa < 0$): the two annotators applied the criterion differently. We use the automatic copy ratio instead.

G.2 Results

Agreement. Annotators agreed on 70.9% (Arabic) and 74.9% (Malay) of Q1 labels (nominal $\kappa = 0.58$ and 0.61). Agreement on the ordinal scale is higher in Arabic (quadratic-weighted $\kappa = 0.80$ vs. 0.68 in Malay), and for the binary supported-versus-not judgment is $\kappa = 0.79$ and 0.65. Correctness agreement (Q2) is high (weighted $\kappa = 0.76$ and 0.87). The largest source of Q1 disagreement in Arabic is that one annotator uses X for empty or garbled answers where the other uses 0. Agreement on the copy label was near zero ($\kappa = -0.05$ and -0.02), so we do not use it.

Collapsed DoRA outputs are unsupported (Arabic) or directionally so (Malay). Under the composite objective the share of DoRA answers judged fully supported falls from 0.578 to 0.085 in Arabic (difference -0.493, 95% CI $[-0.658, -0.328]$) and from 0.427 to 0.273 in Malay (-0.154, $[-0.340, +0.032]$). The Arabic interval excludes zero under the adjudicated labels and under each annotator's labels alone (-0.406 and -0.364, both intervals excluding zero). The Malay difference is negative under all three labelings (-0.154 adjudicated, -0.251 and -0.112 for the two annotators), but its interval excludes zero only for one annotator; we therefore treat the Malay result as consistent with, not independent confirmation of, the collapse.

QLoRA: no human-visible change in this sample. The supported share under the composite objective is essentially unchanged for QLoRA (Arabic $+0.051$, $[-0.167, +0.269]$; Malay $+0.020$, $[-0.234, +0.274]$), and the sign for Malay differs between the two annotators (-0.097 and $+0.029$). The intervals are wide enough to contain both no effect and an effect of the size suggested by the NLI score. Note that, within this seed-42 sample, the mean NLI score of the QLoRA items does not rise under Config C either (Table G1), so the sample is not able to test the 3-seed NLI-score gain. The human data therefore neither support nor contradict a QLoRA faithfulness benefit. In light of Appendix C.1, the DoRA difference is a property of the checkpointed configuration, not of the reward.

The NLI score is only a weak proxy. Across all 175 items per language, the NLI score correlates with the final human Q1 label at Spearman $\rho = 0.30$ ($p < 10^{-4}$) in Arabic and $\rho = 0.16$ ($p = 0.03$) in Malay. The score therefore carries some human-relevant signal, which is enough to detect a collapse as large as DoRA's, but not enough to certify the small differences, such as the QLoRA gain, that depend on it.

Appendix H Full Fine-Tuning Reference (Config D, uncontrolled)

Full fine-tuning (Config D) was run only under the original protocol and was not repeated; it is single-seed (seed 42) and uses a different optimizer (Adafactor), a lower learning rate, and gradient checkpointing relative to Configs B and C (Section 3). Because it was checkpointed and evaluated without `eval()`, its scores are exposed to the same decoding artifact as checkpointed Config C (Appendix C.1) and cannot be interpreted. It supports no comparative claim in this paper. Descriptively, Config D scores $F_{\text{faith}} = 0.0544$ for Arabic and 0.1144 for Malay on mT5, with Arabic exact match 0.019, all below the corresponding Config B and C(off) means of LoRA, QLoRA and DoRA for F_{faith} (Table C8). Replicating Config D under a matched optimizer, evaluation mode and seeds was planned (`baselines` stage) but not run (Section 6).

Appendix I Worked Examples

Figures I3 and I4 show five items drawn at random (seeded draw, no selection) from the 60 per language that were rated by the annotators, among items for which both a Config B and a Config C answer (LoRA, seed 42) were shown. Each card gives the passage, question, reference answer, the two generated answers, the reward-model

Example 1 (Arabic)

Passage

تم العثور على مستحثات أثرية في المرسب الأحفوري يعتقد بأنها لمشطيات ذات أجسام ناعمة وهلامية، تبدو بلا مجسّأت بل لها من أهداب مصطفة أكثر مما في الأشكال الحديثة، وتعود إلى أوائل العصر الكمبري منذ حوالي 515 مليون سنة. لطالما كان موضع المشطيات في تطور شجرة العائلة محلاً للنقاش، وباعتماد علم الوراثة العرفي الجزيئي، فإن وجهة نظر الأغلبية هي أن اللواسع وثنائيات التناظر لهما ارتباط وثيق بعضهما يزيد عن ارتباطهما بالمشطيات. وقد استنتج تحليل وراثي جزيئي أن أصل جميع المشطيات الحديثة كان مشابها للمشطيات، وأن كل المجموعات العصرية وجدت ربّما بعد انقراض العصر الطباشيري الثلاثي منذ 66 مليون سنة خلت. تشير الأدلة التي جُمعَت منذ ثمانينات القرن الماضي إلى أن "المشطيات" ليست أحادية الأرومة، أي لا تتضمن سليلات كاملة ووحيدة لنوع رئيسي مشترك، لأن كل المجموعات الأخرى للمشطيات تنحدر من حيوانات مشطية أخرى

Question

ما الشيء الذي تملكه المشطيات المعاصرة والذي لم يوجد بمستحثات تعود للمشطيات؟

Reference

مجسّأت

	Generated answer (LoRA, seed 42)	NLI (reward)	NLI (indep.)	Supp. A1/A2	Corr. A1/A2
B	515 مليون سنة	0.00	0.04	2 / 2	1 / 1
C	الأشكال الحديثة. وقد استنتج تحليل وراثي جزيئي أن أصل جميع المشطيات الحديثة كان مشابها للمشطيات. وقد استنتج تح	0.46	0.97	1 / 2	1 / 1

Example 2 (Arabic)

Passage

المسائل المحفوظة هي الموضوعات التي هي خارج نطاق السلطة التشريعية لبرلمان اسكتلندا. البرلمان الاسكتلندي غير قادر على إصدار تشريعات بشأن القضايا المحجوزة والتعامل معها، وستمنستر (وحيث تقع الوظائف الوزارية عادةً مع وزراء حكومة المملكة المتحدة). ويشمل ذلك الإجهاض وسياسة البث والخدمة المدنية والأسواق المشتركة للسلع والخدمات في المملكة المتحدة والدستور والكهرباء والفحم والنفط والغاز والطاقة النووية والدفاع والأمن القومي وسياسة الأدوية والتوظيف والسياسة الخارجية والعلاقات مع أوروبا معظم جوانب سلامة النقل والتنظيم واليانصيب الوطني وحماية الحدود والضمان الاجتماعي واستقرار النظام المالي والاقتصادي والنقدي في المملكة المتحدة.

Question

ما هو اسم الأمور خارج القدرة التشريعية للبرلمان الاسكتلندي؟

Reference

المسائل المحفوظة

	Generated answer (LoRA, seed 42)	NLI (reward)	NLI (indep.)	Supp. A1/A2	Corr. A1/A2
B	وستمنستر	0.96	0.05	2 / 2	1 / 1
C	وستمنستر	0.96	0.05	2 / 2	1 / 1

Fig. I3 Randomly drawn worked examples, Arabic (Examples 1–2). NLI (reward) and NLI (indep.): entailment scores from the reward model and from the independent scorer; Supp.: passage support 0–2; Corr.: correctness 0/1; A1/A2: Annotators 1 and 2. B: cross-entropy; C: reward-trained.

and independent NLI scores, and the two annotators' support (0–2) and correctness (0/1) ratings. In Example 1 the reward-trained answer is a long sentence that both NLI models score as more entailed than the short cross-entropy answer, while the first annotator rates its support lower; in Example 2 both conditions give the same answer. The complete set of 248 judgments per annotator with the passages is released with the data.

Example 3 (Arabic)

Passage

كما وجدت الدراسة أنه كان هنالك فرعان حيويان (مجموعات جينية وحيدة الأصل) لم تسبق معرفتهما ولكنهما مرتبطين بجينوم بكتيريا اليرسينيا الطاعونية ذات الصلة بالقبور الجماعية في العصور الوسطى. اتضح أن هذين الفرعين الحيويين (الذين يعتقد أنهما انقرضا) من أسلاف المستفردات الحديثة لسلالات اليرسينيا الطاعونية الحديثة يرسينيا بيستيس أوريانتاليس ويرسينيا بيستيس ميديفاليس، مما يشير إلى أن الطاعون قد يكون قد دخل أوروبا في موجتين. تشير عمليات المسح لبقايا الطاعون في فرنسا وإنجلترا إلى أن المتغير الأول دخل أوروبا عبر ميناء مارسيليا حوالي العام 1347 وانتشر في أرجاء فرنسا على مدى العامين التاليين، فوصل إنجلترا في النهاية في ربيع العام 1349 حيث تفشى في جميع أنحاء البلاد في ثلاثة أوبئة. أظهرت عمليات المسح لبقايا الطاعون من بلدة بيرخن أوب زووم الهولندية أن التركيب الوراثي يرسينيا بيستيس المسؤول عن الوباء الذي انتشر عبر البلدان المنخفضة منذ عام 1350 كان يختلف عن ذلك الموجود في بريطانيا وفرنسا، مما يشير إلى أن بيرخن أوب زووم (وأجزاء أخرى محتملة من جنوب هولندا) لم تصب بالعدوى مباشرة من إنجلترا أو فرنسا في عام 1349، مما يشير إلى احتمال حدوث موجة ثانية من الطاعون مختلفة عن تلك الموجودة في بريطانيا وفرنسا قد انتقلت إلى البلدان المنخفضة من النرويج أو المدن الهانزية أو موقعٍ آخر.

Question

ما الذي تشير إليه سلالات يرسينيا الطاعونية بشأن الطاعون؟

Reference

الطاعون قد يكون قد دخل أوروبا في موجتين

	Generated answer (LoRA, seed 42)	NLI (reward)	NLI (indep.)	Supp. A1/A2	Corr. A1/A2
B	فرنسا وفرنسا	0.63	0.22	1 / 2	1 / 1
C	ثلاثة أوبئة. تشير عمليات المسح لبقايا الطاعون في فرنسا وفرنسا	1.00	0.44	1 / 2	1 / 1

Example 4 (Malay)

Passage

Para saintis telah menemui bentuk muka bumi yang berselarak pada permukaan bulan yang dipanggil tubir lobat yang disebabkan oleh bulan yang mengecut dengan amat perlahan. Serpihan ini telah ditemui di serata permukaan bulan dan masih kurang terjejas oleh faktor persekitaran, yakni secara geologinya menunjukkan bahan tersebut adalah baharu terbentuk. Teori ini bercanggah dengan dakwaan bahawa bulan adalah sepenuhnya tanpa aktiviti geologi.

Question

Menurut petikan, teori manakah berkaitan pembentukan permukaan yang menandakan bahawa terdapat aktiviti geologi yang berlaku di bulan?

Reference

Pembentukan permukaan dicipta dalam tempoh yang agak baru

	Generated answer (LoRA, seed 42)	NLI (reward)	NLI (indep.)	Supp. A1/A2	Corr. A1/A2
B	Pembentukan permukaan dicipta sejak dahulu	0.00	0.00	0 / 1	0 / 1
C	Pembentukan permukaan dicipta sejak dahulu	0.00	0.00	0 / 1	0 / 1

Example 5 (Malay)

Passage

Paralimpik akan berlangsung dari 24hb Ogos hingga 5hb September 2021. Sebilangan acara akan diadakan di lokasi lain di seluruh Jepun. Tokyo akan menjadi satu-satunya bandar Asia untuk menjadi tuan rumah Olimpik musim panas dua kali, setelah pernah menjadi tuan rumah pada tahun 1964. Jika anda menempah penerbangan dan penginapan anda pada 2020 sebelum penangguhan diumumkan, anda mungkin menghadapi situasi yang sukar. Polisi pembatalan berbeza-beza, tetapi pada akhir bulan Mac kebanyakan dasar pembatalan berasaskan korfonavirus tidak berlaku sehingga Julai 2020 ketika Olimpik telah dijadualkan. Adalah dijangka tiket peristiwa sebahagian besar akan berharga di antara ¥2,500 dan ¥130,000, dengan biasa tiket berharga sekitar ¥7,000.

Question

Bilakah Olimpik musim panas dijadualkan berlangsung pada asalnya?

Reference

Julai 2020

	Generated answer (LoRA, seed 42)	NLI (reward)	NLI (indep.)	Supp. A1/A2	Corr. A1/A2
B	Mac 2020	0.02	0.02	1 / 2	1 / 1
C	September 2021	0.01	0.01	2 / 2	0 / 1

Fig. I4 Randomly drawn worked examples, Arabic (Example 3) and Malay (Examples 4–5); columns as in Figure I3.

Appendix J Supplementary Tables for the Primary Study

Table J2 Paired contrasts of F1 and exact match (Ar: Arabic, Ma: Malay, T5G2: T5Gemma 2; mean difference with 95% bootstrap interval over held-out items; *: interval excludes zero; no multiplicity adjustment).

Cell	Contrast	F1	EM
Qwen3-Ar	C−B	+.011* [+.001,+.021]	+.009 [−.004,+.022]
Qwen3-Ar	C−$C_{\lambda=0}$	+.013* [+.002,+.023]	+.007 [−.006,+.019]
Qwen3-Ar	C−C_{shuf}	+.004 [−.006,+.015]	−.003 [−.016,+.009]
Qwen3-Ma	C−B	−.021* [−.040,−.003]	−.021 [−.043,.000]
Qwen3-Ma	C−$C_{\lambda=0}$	−.016 [−.034,+.002]	−.015 [−.037,+.006]
Qwen3-Ma	C−C_{shuf}	−.019 [−.039,.000]	−.021 [−.043,.000]
T5G2-Ar	C−B	+.004 [−.009,+.016]	−.004 [−.017,+.009]
T5G2-Ar	C−$C_{\lambda=0}$	−.006 [−.018,+.005]	−.006 [−.019,+.007]
T5G2-Ar	C−C_{shuf}	−.006 [−.017,+.007]	−.003 [−.016,+.010]
T5G2-Ma	C−B	.000 [−.017,+.017]	+.001 [−.019,+.021]
T5G2-Ma	C−$C_{\lambda=0}$	+.009 [−.010,+.028]	+.013 [−.008,+.034]
T5G2-Ma	C−C_{shuf}	−.011 [−.032,+.008]	−.008 [−.031,+.014]

Table J3 Mean NLI score (number of answers) by answer length in words, all seeds pooled.

Condition	1	2–3	4–6	7–10	11+
Qwen3-0.6B-Base, Arabic					
A (frozen)	0.07 (62)	0.04 (57)	0.32 (31)	0.70 (104)	0.59 (466)
B (CE)	0.07 (814)	0.11 (1730)	0.21 (779)	0.33 (218)	0.60 (59)
C (CE+NLI)	0.07 (765)	0.11 (1729)	0.20 (750)	0.44 (238)	0.53 (118)
C, λ=0	0.08 (799)	0.10 (1759)	0.20 (772)	0.39 (199)	0.64 (71)
C, shuffled	0.07 (834)	0.11 (1731)	0.22 (770)	0.41 (195)	0.52 (70)
Qwen3-0.6B-Base, Malay					
A (frozen)	0.05 (28)	0.07 (84)	0.33 (151)	0.39 (108)	0.44 (59)
B (CE)	0.07 (279)	0.20 (775)	0.41 (707)	0.40 (333)	0.67 (56)
C (CE+NLI)	0.06 (279)	0.20 (795)	0.42 (680)	0.44 (350)	0.61 (46)
C, λ=0	0.07 (289)	0.20 (776)	0.40 (700)	0.43 (332)	0.61 (53)
C, shuffled	0.06 (280)	0.21 (797)	0.41 (699)	0.40 (323)	0.67 (51)
T5Gemma 2, Arabic					
A (frozen)	0.03 (176)	0.01 (13)	0.01 (5)	0.12 (40)	0.25 (486)
B (CE)	0.09 (847)	0.10 (1840)	0.20 (716)	0.37 (159)	0.46 (38)
C (CE+NLI)	0.08 (801)	0.10 (1933)	0.21 (705)	0.41 (123)	0.40 (38)
C, λ=0	0.09 (790)	0.10 (1926)	0.19 (716)	0.39 (140)	0.44 (28)
C, shuffled	0.10 (789)	0.10 (1830)	0.19 (751)	0.34 (171)	0.42 (59)
T5Gemma 2, Malay					
A (frozen)	0.00 (37)	0.01 (10)	–	0.00 (5)	0.10 (378)
B (CE)	0.06 (260)	0.20 (837)	0.43 (681)	0.39 (321)	0.62 (51)
C (CE+NLI)	0.07 (250)	0.21 (821)	0.41 (667)	0.43 (359)	0.61 (53)
C, λ=0	0.07 (267)	0.20 (830)	0.41 (666)	0.42 (340)	0.55 (47)
C, shuffled	0.06 (251)	0.20 (836)	0.40 (672)	0.45 (344)	0.60 (47)

Table J4 Repeated runs of the same seed and condition (Qwen3, condition $C_{\lambda=0}$). The two executions used identical code, data and seed on different processes; which of them is stored in the released run record is not recorded. Values are from the training logs (three decimals).

		Execution 1			Execution 2		
Language	**Seed**	Faith	F1	EM	Faith	F1	EM
Arabic	2026	0.149	0.605	0.454	0.145	0.591	0.432
Arabic	99	0.154	0.603	0.435	0.151	0.612	0.436
Malay	42	0.292	0.625	0.563	0.303	0.646	0.584
Malay	99	0.280	0.635	0.558	0.292	0.628	0.556

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- **Use of AI-assisted tools:** Disclosed in Section 3.8.